Band Equivalence, Hopping Locality, and Internal Fidelity of Reduced Wannier Hamiltonians

From Exact Rank-Two Geometry to a DFT-Derived Rank-10 Silicon Case Study

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Abstract

Band energies are invariant under momentum-dependent unitary similarity, whereas a Hamiltonian's matrix representation and real-space hopping locality are not. We combine an exact rank-two theory of this distinction with a source-relative case study of a DFT-derived rank-10 scalar-relativistic silicon Wannier Hamiltonian. In the controlled rank-two family, spectral and globally aligned operator losses have an exact common-curvature geometry. For $U_a(q) = R(a \sin 2\pi q)$, the minimum restricted-alignment threshold is $\sqrt{\kappa[1 - J_0(2a)^2]}$; for a flat traceless parent, its square equals the normalized Fourier power transferred from the onsite block to nonzero displacement. A dispersive development calculation preserves complete bands while changing the first sampled cutoff meeting a $10^{-8}E_{\mathrm{ref}}$ band-error target from 4 to 10.

The silicon study uses source files whose provenance, byte identities, native optimizer completion, Hermiticity, pair closure, and reconstruction were verified; this "authenticated" status is not material validation. A frozen internal gauge attack preserves the complete represented spectrum to 3.20×10^{-14} eV while changing restricted alignment and finite-block locality. Frozen shell-reduced models then show that the 408-point fitting inventory understates exact Parseval operator loss and dense-mesh spectral loss, and that global spectral RMS does not control conduction curvature or the nearly degenerate valence manifold. A polar-transported three-band quadratic Hamiltonian is therefore used instead of independently ordered hole tensors.

The source fails separate DFT-parent gap and electron-mass targets, and the silicon results remain producer-checked development evidence pending independent reconstruction. Within those boundaries, the results establish that spectrum, represented operator, hopping locality, interpolation fidelity, and band-edge fidelity are distinct reduction objectives; near-full native support is required to preserve them simultaneously for the studied source.

Keywords: band equivalence; effective mass; hopping locality; model reduction; silicon; Wannier Hamiltonian

1 Introduction

First-principles electronic-structure calculations produce operators and states in representations designed for numerical convergence rather than for compact real-space modeling. Plane waves are systematic and broadly transferable, but they are expensive for repeated interpolation, large supercells, local perturbations, and reduced-device calculations. Wannier representations replace a selected Bloch subspace by localized coordinates and expose a Fourier hierarchy of matrix-valued hopping blocks [1, 2, 3, 4, 5]. A further reduced model may then retain fewer blocks, impose symmetry patterns, or fit a smaller parameter family.

The apparent objective—"reproduce the bands with a shorter Hamiltonian"—contains several different mathematical tasks. Band energies constrain eigenvalues. An operator comparison

constrains matrices after identifying their state spaces. A locality comparison constrains Fourier coefficients in a declared basis and gauge. Interpolation asks how a frozen model behaves away from the points used to characterize it. Effective masses and valley locations depend on local derivatives and extrema. These requirements are related, but none is interchangeable with all the others.

The gauge freedom of a retained Bloch manifold makes the distinction unavoidable. If $H(\mathbf{k})$ is transformed by a momentum-dependent unitary $U(\mathbf{k})$,

$$\widetilde{H}(\mathbf{k}) = U(\mathbf{k})^{\dagger} H(\mathbf{k}) U(\mathbf{k}), \tag{1}$$

then the complete eigenvalue set is unchanged at every momentum. The matrix field and its Fourier coefficients generally change. A momentum-independent unitary merely conjugates each real-space block; a momentum-dependent unitary convolves displacement indices and can redistribute weight among displacement shells, including toward longer range for particular transformations and locality measures. No universal monotonic increase is implied. Band equivalence therefore does not determine hopping locality.

This paper develops that statement through two complementary but logically distinct levels. The first is a controlled theoretical result and synthetic benchmark. The second is a source-specific computational case study; it is not an empirical proof of a universal rank-N extension of the rank-two identities.

1. A controlled one-dimensional rank-two problem supplies exact loss geometry, Bessel-function threshold laws, a residual–Fourier-power identity, and a dispersive fixed-accuracy range-inflation calculation.

2. A DFT-derived rank-10 silicon Wannier Hamiltonian tests the same distinctions in a higher-rank, three-dimensional represented field with realistic matrix structure, native Wigner–Seitz blocks, dense interpolation, conduction valleys, and a degenerate valence edge.

The silicon calculation is not the "validation" of the synthetic theorem in the sense of proving production material accuracy. It is the next rung of a validation ladder: the exact mechanism is first isolated in a known model, then deployed on an authenticated material-derived operator while keeping parent-model, representation, conversion, truncation, fitting, and feature errors separate.

The main questions are:

1. How can complete spectra remain identical while restricted alignment and real-space locality change?

2. What exact relationship connects irreducible matrix mismatch to induced hopping weight in a controlled rank-two family?

3. How does representation-dependent locality affect the finite range needed at equal accuracy?

4. How much native support is required to compress the rank-10 silicon Wannier source by literal shell deletion or bounded shell refitting?

5. Do errors measured on a finite 408-point inventory predict exact integrated operator error, dense interpolation error, and band-edge fidelity?

6. What is the appropriate hole-sector object when the non-SOC valence edge is a nearly degenerate three-band manifold?

The answers are deliberately bounded. This article is a long-form synthesis of the exact rank-two framework and the completed rank-10 silicon calculation program. Analytic statements are identified separately from sampled numerical observations. The active rank-two numerical headline

remains development or partial evidence; its package has not yet been finalized as an immutable archive. The non-flat numerical results are sampled on declared inventories. The silicon gauge, shell, fidelity, and band-edge studies are authenticated producer-checked development analyses that have not been independently reconstructed. They are internal to a source that fails separate DFT-parent adequacy targets and do not validate the source as an accurate silicon Hamiltonian. These boundaries are not qualifications added at the end; they define the scientific question from the beginning. Preparation of this working manuscript included one separately authorized fixed-density Quantum ESPRESSO parent band-path calculation after the reduction models had been frozen; it changed no source construction, fit, selection, or threshold. No Wannier90 execution was performed, and manuscript preparation does not itself authorize submission, publication, or release.

2 Background and conceptual framework

2.1 Physical model, represented operator, and finite matrix

A physical electronic-structure model, an operator on a retained state space, and a finite matrix representation are different objects. The Kohn–Sham calculation defines an approximation to the physical parent under a chosen functional, pseudopotential, geometry, and numerical setting [6, 7, 8, 9]. Wannierization selects and represents a finite subspace. A reduced hopping model then approximates that represented operator. Errors introduced at these stages must not be collapsed into one number unless a reviewed contract explicitly defines the combination.

For a finite-rank periodic representation,

$$H(\mathbf{k}) = \sum_{\mathbf{R}} e^{i\mathbf{k}\cdot\mathbf{R}} T(\mathbf{R}), \tag{2}$$

where $T(\mathbf{R})$ is a matrix block associated with a lattice translation. Equation (2) is meaningful only after the lattice basis, reciprocal coordinates, representative convention, degeneracies, units, basis ordering, and energy reference are fixed.

The silicon program uses two Fourier routes. Native Wannier90 HR data use Wigner–Seitz representatives and degeneracy factors. The generic gauge study samples a matrix field on a centered finite grid, performs a finite-cyclic transform, and then converts explicitly to ordinary lattice representatives. These routes answer related questions but are not interchangeable. Their conversion and interpolation consequences are retained as separate channels.

2.2 Spectral reduction and finite representation extent

The distinction between representation and reduction can be seen most cleanly in two retained one-dimensional calibration problems. In the particle-in-a-box calculation, a continuum differential operator, its finite-difference matrix, the computed matrix eigenpairs, and a low-energy spectral compression are separate objects. An eigensolver may solve the finite matrix to high accuracy while that matrix still approximates the continuum spectrum poorly. If the retained vectors are exact eigenvectors of the finite matrix, however, compression to their span is algebraically exact for that represented matrix. With

$$P_M = \sum_{n=1}^{M} |u_n\rangle\langle u_n|, \qquad Q_M = I - P_M, \qquad H_M = P_M H P_M, \tag{3}$$

one has $Q_M H P_M = 0$ for an invariant spectral subspace [10, 11, 12]. This identity establishes closure of the reduction; it does not establish that the finite parent represents the continuum or a material accurately. For a non-invariant trial space, leakage $Q_M H P_M$ becomes a separate error channel. Entangled-band Wannier disentanglement is an instance of this more general distinction: the rank-10 silicon trial subspace is selected from 20 parent bands and is not globally an invariant

spectral subspace of that parent. At construction nodes, eigenstates inside the frozen inner window are included exactly by the declared disentanglement constraint; away from those nodes, the selected subspace and its interpolation determine derivative fidelity. Exact frozen-window inclusion therefore does not by itself control interpolated curvatures or leakage near the window boundary [2, 3, 13].

The harmonic-oscillator calibration introduces a second limitation: finite spatial extent. A real-line operator represented on a finite Dirichlet interval has changed its domain and boundary realization before the grid is refined. Decreasing the grid spacing at fixed interval width resolves that finite-box problem more accurately; it does not by itself recover the real-line operator. Boundary truncation, interior discretization, retained spectral dimension, coordinate injection, eigensolver error, and floating-point error therefore remain distinct [14]. This is the same logical reason that agreement inside a reduced model cannot repair an inadequate parent.

For the silicon hierarchy, four notions of "finite" must consequently not be merged:

1. a finite plane-wave cutoff and reciprocal sampling define a numerical DFT representation, even though each plane wave extends across the cell;
2. a finite band/Wannier rank defines a retained spectral state space;
3. a finite real-space hopping support defines the range of the represented lattice Hamiltonian; and
4. a finite evaluation inventory defines only the reciprocal points on which a loss has been sampled.

Taking the plane-wave cutoff, construction mesh, retained rank, hopping radius, or evaluation mesh to a larger value addresses a different approximation. In particular, restoring all native hopping blocks converges to the fixed Wannier source, not to its DFT parent; refining a support-limited model cannot remove a source/parent discrepancy. This nested-limit view is the conceptual bridge from the one-dimensional particle-in-a-box (PIAB1D) and quantum-harmonic-oscillator (QHO1D) calibration reports to the present periodic reduction study.

2.3 Gauge freedom and locality

Wannier functions inherit the gauge freedom of a retained Bloch subspace [1, 2, 3]. Smooth gauges can differ in spread and real-space shape even when they span the same projector. Here let $\mathbf{m}$, $\mathbf{n}$, and $\mathbf{Q}$ denote physical direct-lattice translation vectors, in the same convention as $\mathbf{R}$ in Eq. (2); they are not the reduced integer coordinates used later in the finite 6^3 transform. For $U(\mathbf{k}) = \sum_{\mathbf{m}} e^{i\mathbf{k}\cdot\mathbf{m}} U(\mathbf{m})$, substitution into Eq. (1) gives the hopping convolution

$$\widetilde{T}(\mathbf{Q}) = \sum_{\mathbf{m},\mathbf{n}} U(\mathbf{m})^{\dagger} T(\mathbf{Q} + \mathbf{m} - \mathbf{n}) U(\mathbf{n}). \tag{4}$$

When reduced integer coordinates are used instead, the physical phase $\mathbf{k} \cdot \mathbf{R}$ becomes $2\pi\boldsymbol{\kappa} \cdot \mathbf{m}$ explicitly. A constant transformation has only one Fourier coefficient and does not enlarge support. A momentum-dependent transformation mixes displacement indices. Locality is therefore a property of the represented field and allowed gauge, not of the complete spectrum alone.

This does not make locality arbitrary. Once the construction, retained space, gauge, metric, support family, and error criterion are fixed, the resulting locality problem is well defined. The important restriction is epistemic: spectral data alone cannot identify a unique hopping distribution.

2.4 Gauge invariants, alignment families, and identifiability

For each momentum, unitary similarity preserves the characteristic polynomial, trace invariants, and the complete unordered spectrum:

$$\det[zI - U(\mathbf{k})^\dagger H(\mathbf{k})U(\mathbf{k})] = \det[zI - H(\mathbf{k})]. \tag{5}$$

It also preserves every pointwise unitarily invariant norm of the complete operator. It does not preserve individual matrix entries, basis-resolved observables, or Fourier coefficients. Consequently, an eigenvalue-only inverse problem has an entire gauge orbit of solutions. A locality statement becomes identifiable only after the allowed coordinate transformations are restricted.

The present work distinguishes four such restrictions. Identity alignment compares matrices in the declared common Wannier basis. Global alignment selects one momentum-independent right unitary from the represented frames by the stacked Procrustes problem

$$Q_{\mathrm{global}} = \arg\min_{Q\in U(N)} \sum_j w_j \, \|X(\mathbf{k}_j)Q - Y(\mathbf{k}_j)\|_F^2 \,, \tag{6}$$

and subsequently compares the matrices after conjugation by that selected Q. This is a frame-alignment prescription with a closed polar-factor solution; it is not claimed to solve the generally different quartic minimization of a matrix-conjugation residual. Pointwise polar alignment admits a different complete unitary at every point. If two orthonormal frame matrices X and Y have overlap $X^\dagger Y = L\Sigma R^\dagger$, the polar/Procrustes action is $Q = LR^\dagger$ when the declared singular-value rank gate is satisfied [15, 16]. Finally, a known-inverse action uses the unitary supplied by a synthetic construction. The latter two are oracle controls: they check the representation and implementation, but they are not available from the complete spectra alone.

The matrix residual after Eq. (6) is conditional on this particular frame-derived global-unitary prescription; it is not asserted to be an invariant metric between unrestricted gauge orbits or the global optimum of matrix conjugation. A nonzero identity or global residual is therefore not evidence that two complete unitarily related fields represent different physics. It states that the generating momentum dependence lies outside the admitted alignment family. Conversely, allowing an independently fitted pointwise unitary in a same-gauge truncation study would erase part of the approximation being measured. This is why the shell-reduction operator loss uses identity alignment.

2.5 Norm relations, Parseval identity, and support truncation

For Hermitian matrices A and B, the ordered eigenvalues satisfy the Hoffman–Wielandt bound

$$\sum_{n=1}^{N} [\lambda_n(A) - \lambda_n(B)]^2 \le \|A - B\|_F^2 \,. \tag{7}$$

Thus an operator residual controls an ordered spectral residual at the same momentum. The converse is false: isospectral matrices can have a finite Frobenius separation. This one-way implication motivates retaining both loss coordinates rather than treating one as a surrogate for the other.

For two matrix fields in the same Fourier convention, $\Delta H(\mathbf{k}) = \sum_{\mathbf{R}} e^{i\mathbf{k}\cdot\mathbf{R}} \Delta T(\mathbf{R})$, normalized Brillouin-zone integration gives the matrix-valued Parseval identity

$$\frac{1}{\Omega_{\mathrm{BZ}}} \int_{\mathrm{BZ}} \|\Delta H(\mathbf{k})\|_F^2 \, d\mathbf{k} = \sum_{\mathbf{R}} \|\Delta T(\mathbf{R})\|_F^2 \,. \tag{8}$$

For rank N and reporting energy E_0, the exact integrated per-state operator RMS is

$$\tau_{2,\mathrm{BZ}} = \frac{1}{E_0} \left[\frac{1}{N} \sum_{\mathbf{R}} \|\Delta T(\mathbf{R})\|_F^2 \right]^{1/2} . \tag{9}$$

No reciprocal sampling is required once the unique coefficient convention is fixed. This exact identity is used in the fidelity ladder.

For a literal support S, the truncated model is

$$H_S(\mathbf{k}) = \sum_{\mathbf{R}\in S} e^{i\mathbf{k}\cdot\mathbf{R}} T(\mathbf{R}), \qquad \Delta T_S(\mathbf{R}) = \begin{cases} 0, & \mathbf{R} \in S, \\ -T(\mathbf{R}), & \mathbf{R} \notin S. \end{cases} \tag{10}$$

Equation (9) then makes the integrated operator loss exactly the excluded Frobenius coefficient power. A separate uniform operator bound follows from the triangle inequality,

$$\sup_{\mathbf{k}} \|H_S(\mathbf{k}) - H(\mathbf{k})\|_2 \le \sum_{\mathbf{R}\notin S} \|T(\mathbf{R})\|_2 \,. \tag{11}$$

The Frobenius tail and the spectral-norm triangle bound answer different questions: the first is an exact integrated RMS quantity, while the second is generally a conservative uniform bound. Neither determines the local curvature of an individual band.

2.6 Spectral and operator losses

For a reduced model $H_{\mathrm{r}}(\mathbf{k})$ in the same basis as a reference $H_{\mathrm{W}}(\mathbf{k})$, a spectral loss compares ordered eigenvalues, while an operator loss compares matrices. On the silicon 408-point inventory the primary normalized RMS coordinates are

$$\tau_1^2 = \frac{1}{4080} \sum_{\mathbf{k},n} \left(\frac{\epsilon^{\mathrm{r}}_{n\mathbf{k}} - \epsilon^{\mathrm{W}}_{n\mathbf{k}}}{1\,\mathrm{eV}} \right)^2 , \tag{12}$$

$$\tau_2^2 = \frac{1}{4080} \sum_{\mathbf{k}} \frac{\|H_{\mathrm{r}}(\mathbf{k}) - H_{\mathrm{W}}(\mathbf{k})\|_F^2}{(1\,\mathrm{eV})^2}. \tag{13}$$

Here $4080 = N_k N = 408 \times 10$, so both quantities are per reciprocal point and per retained state. The 1 eV denominator is a reporting normalization, not an acceptance limit. Identity alignment is used for same-gauge shell models because a fitted pointwise alignment would erase part of the reduction error.

A matrix perturbation bounds spectral displacement, but spectral agreement does not recover the matrix. Consequently, (τ_1, τ_2) is treated as a two-dimensional loss coordinate. Post-hoc threshold maps summarize observed witnesses; they do not become prospective scientific tolerances after the fact.

2.7 Global averages and band-edge observables

A global RMS loss averages across bands and reciprocal points. A band edge depends on a local extremum and its derivatives. For a nondegenerate band,

$$(m^{-1})_{ij} = \frac{1}{\hbar^2} \frac{\partial^2 E}{\partial k_i \partial k_j}. \tag{14}$$

Differentiation can amplify small local shape errors that contribute little to a global average. Valley displacement, Hessian rotation, and principal masses are therefore first-class reduction outputs rather than decorations on a band plot.

At a degenerate valence edge, independently ordered branches may exchange identity and fail to be analytic. A single "hole mass tensor" is then not basis invariant. The appropriate object is a coupled matrix-valued quadratic Hamiltonian, consistent with multiband effective-mass reasoning [17]. In the present non-SOC source, this object describes three orbital valence channels; it is not assigned physical heavy-hole, light-hole, or split-off labels.

Let P_0 project onto an r-dimensional cluster at $\mathbf{k}_0$. A smooth nearby cluster with projector $P(\mathbf{q})$ must first be identified with $P_0\mathbb{C}^N$ before matrices can be subtracted. If X_0 and $X(\mathbf{q})$ are orthonormal cluster frames, the singular values of $X_0^\dagger X(\mathbf{q})$ are cosines of principal angles [18]. The polar factor of this overlap supplies a least-squares unitary identification whenever the overlap remains full rank. The transported represented cluster Hamiltonian can then be expanded as

$$H_{\mathrm{eff}}(\mathbf{q}) = H_{\mathrm{eff}}(0) + \sum_i q_i A_i + \sum_{i,j} q_i q_j B_{ij} + O(\|\mathbf{q}\|^3). \tag{15}$$

A symmetric $+\mathbf{q}/-\mathbf{q}$ average removes the odd term without assigning individual branch identities. For a unit direction $\hat{\mathbf{n}}$, the eigenvalues of $2\sum_{ij} n_i n_j B_{ij}$ are the quadratic curvature channels. These channels are basis invariant under one common unitary rotation of the cluster coefficient field. They need not equal Hessians of independently ordered bands when $H_{\mathrm{eff}}(0)$ is degenerate or when a reduction has artificially split it.

3 Exact rank-two theory

3.1 Trace–traceless candidate family

Write a rank-two represented Hamiltonian as

$$H(q) = \bar{E}(q) I_2 + D(q), \qquad \operatorname{tr} D(q) = 0. \tag{16}$$

The candidate family shifts the centroid and positively rescales the splitting,

$$C_{s,\lambda}(q) = \bar{E}(q) I_2 + s E_{\mathrm{ref}} I_2 + \lambda D(q), \qquad \lambda > 0. \tag{17}$$

The comparison target is a pointwise band-equivalent field

$$A_a(q) = U_a(q)^\dagger H(q) U_a(q), \qquad U_a(q) = R(a \sin 2\pi q). \tag{18}$$

The admissible alignment is one momentum-independent real rotation $R(\phi)$, not the generating momentum-dependent map.

With positive normalized weights α_i, define

$$\kappa = \sum_i \frac{\alpha_i}{2E_{\mathrm{ref}}^2} \|D(q_i)\|_F^2 . \tag{19}$$

The losses use the same positive weights, normalized by $\sum_i \alpha_i = 1$:

$$L_{\mathrm{spec}}^2 = \sum_i \frac{\alpha_i}{2E_{\mathrm{ref}}^2} \sum_{n=1}^{2} [\epsilon_n(C_{s,\lambda}; q_i) - \epsilon_n(A_a; q_i)]^2, \tag{20}$$

$$L_{\mathrm{op},\phi}^2 = \sum_i \frac{\alpha_i}{2E_{\mathrm{ref}}^2} \left\| R(\phi)^{\mathsf{T}} C_{s,\lambda}(q_i) R(\phi) - A_a(q_i) \right\|_F^2 . \tag{21}$$

For an allowed global angle, let μ_ϕ be the normalized weighted correlation between the globally rotated candidate traceless field and the attacked target. A real rotation conjugates the two-dimensional traceless symmetric coordinates through angle 2ϕ. Consequently the correlation has the explicit form

$$\kappa\mu_\phi = b_0 + c\cos 2\phi + d \sin 2\phi, \qquad \kappa\mu_{\max} = b_0 + \sqrt{c^2 + d^2}. \tag{22}$$

This is why the global alignment reduces to one second harmonic rather than a numerical search over an unrestricted field of rotations. Writing $D = x\sigma_x + y\sigma_y + z\sigma_z$, real rotations leave the σ_y coefficient unchanged, so $b_0 = \sum_i \alpha_i y(q_i)^2 / E_{\mathrm{ref}}^2 \geq 0$ and hence $\mu_{\max} \geq 0$. The positivity constraint on λ therefore does not clip the reported optimum; if a different candidate orbit allowed $\mu_{\max} < 0$, the constrained minimum would instead occur at $\lambda = 0$.

Theorem 1 (Common-curvature loss geometry)**.** *Under the family above,*

$$L^2_{\mathrm{spec}}(s,\lambda) = s^2 + \kappa(\lambda-1)^2, \tag{23}$$

$$L^2_{\mathrm{op},*}(s,\lambda) = s^2 + \kappa(\lambda-\mu_{\max})^2 + \kappa(1-\mu^2_{\max}). \tag{24}$$

The two losses have common curvature $\mathrm{diag}(1,\kappa)$*. Their full unclipped admissible sets are homothetic ellipses with different centers and, for the operator channel, an irreducible floor* $\kappa(1-\mu^2_{\max})$*.*

The theorem makes the compatibility question explicit. Spectral adequacy centers the splitting scale at $\lambda = 1$; restricted operator adequacy centers it at $\lambda = \mu_{\max}$. The momentum-dependent part that cannot be represented by one global rotation remains as the irreducible floor.

If the spectral and operator thresholds are τ_s and τ_o, define

$$r_*^2 = \tau_o^2 - \kappa(1-\mu^2_{\max}). \tag{25}$$

When the complete threshold ellipses are contained in the declared parameter rectangle, they intersect if and only if

$$\sqrt{\kappa}\,|1-\mu_{\max}| \le \tau_s + r_*. \tag{26}$$

The containment condition is necessary because Eq. (26) compares the full ellipses through their center separation and radii. If the admissible rectangle clips either ellipse, two full ellipses may intersect only outside the allowed parameter domain, or clipping may remove the nearest points used by the center-distance argument. A failed numerical search is therefore not a separation proof; Eq. (26) supplies the analytic decision only together with the containment premises.

3.2 Smooth threshold continuum

For a flat real-symmetric traceless parent, the smooth attack in Eq. (18) yields

$$\mu_{\max}(a) = \frac{1}{2\pi}\int_0^{2\pi} \cos[2a\sin\theta]\,d\theta = J_0(2a), \tag{27}$$

where J_0 is the Bessel function of the first kind [19]. Define the attack angle $\theta_a(q) = a\sin(2\pi q)$. The factor two appears because conjugation by a real rotation through $\theta_a(q)$ rotates the traceless symmetric coordinates through $2\theta_a(q)$. Therefore

$$\tau_{o,\min}(a) = \sqrt{\kappa[1-J_0(2a)^2]}, \tag{28}$$

$$\tau_{o,\mathrm{joint}}(a;0) = \sqrt{2\kappa[1-J_0(2a)]}. \tag{29}$$

On $0 \le a \le \pi/4$, the Bessel argument satisfies $0 \le 2a \le \pi/2$, before the first positive zero $j_{0,1}$ of J_0, so $J_0(2a) > 0$. It is also before the first positive zero of J_1; since $J_0'(x) = -J_1(x) < 0$ on this interval, $J_0(2a)$ decreases and both thresholds are nondecreasing. Outside this interval a negative $J_0(2a)$ would be absorbed by a global $\pi/2$ real rotation, giving the maximal correlation $|J_0(2a)|$ rather than a negative optimal scale. Every member of the family has exactly the same complete bands; only the restricted representation threshold changes.

3.3 Residual–locality identity

Let the flat traceless target after attack be

$$B(q) = U(q)^\dagger D U(q) = \sum_R e^{2\pi i q R} B_R. \tag{30}$$

Theorem 2 (Rank-two residual–non-onsite-power identity)**.** *Let D be flat, traceless, and real symmetric, let $U(q)$ belong to the same real rotation orbit, and use either normalized continuum measure or a finite uniform mesh with $\alpha_i = 1/N$. Then*

$$\inf_{s,\lambda\geq 0,\phi} L_{\mathrm{op},\phi}^2 = \kappa(1-\mu_{\max}^2) = \frac{1}{2E_{\mathrm{ref}}^2}\sum_{R\neq 0}\|B_R\|_F^2\,. \tag{31}$$

The finite statement uses the corresponding uniform discrete Fourier transform, so its optimal constant is exactly the zero Fourier block.

To expose the intermediate step, normalized Parseval gives

$$\sum_{R\neq 0}\|B_R\|_F^2 = \int_{-1/2}^{1/2}\|B(q)\|_F^2\,dq - \|B_0\|_F^2\,. \tag{32}$$

The best constant real rotation and positive scale reproduce the optimally oriented onsite component; the normalized squared onsite amplitude is $\mu_{\max}^2$. Because unitary conjugation preserves $\|B(q)\|_F = \|D\|_F$, the remainder is exactly $\|D\|_F^2\,(1-\mu_{\max}^2)$, yielding Eq. (31). This is the exact bridge between restricted matrix mismatch and hopping locality under the theorem's assumptions. The global rotation can optimize the onsite orientation, but it cannot remove Fourier weight at nonzero displacement. For the smooth family, the threshold in Eq. (28) is the square root of the normalized induced non-onsite power.

3.4 What the theorem does not claim

The result does not establish a universal relation between any matrix residual and any locality measure. It assumes a flat traceless rank-two parent and matched real rotation orbits. A dispersive parent adds native nonzero-range blocks and cross terms. Higher rank introduces noncommuting rotations and a richer gauge group. The value of the controlled theorem is that it isolates a mechanism that can then be tested without silently extending the equality beyond its assumptions.

4 Controlled rank-two calculation

4.1 Finite-range four-state parent

The dispersive benchmark is a one-dimensional periodic model with lattice period a_0 and reduced momentum $q = ka_0/(2\pi) \in [-1/2, 1/2)$. Four states per cell are partitioned into lower and upper two-state sectors. The bounded parent homotopy is controlled by $\rho \in [0,1]$. Its only nonzero hopping blocks are

$$T_0 = \begin{pmatrix} D_{\mathrm{L}} & 0 \\ 0 & D_{\mathrm{U}} \end{pmatrix}, \qquad T_{+1}(\rho) = \rho\begin{pmatrix} J_{\mathrm{L}} & V \\ 0 & J_{\mathrm{U}} \end{pmatrix}, \qquad T_{-1}(\rho) = T_{+1}(\rho)^\dagger, \tag{33}$$

with, in units of E_{ref},

$$D_{\mathrm{L}} = \begin{pmatrix} -1.20 & 0.06 \\ 0.06 & -0.35 \end{pmatrix}, \qquad D_{\mathrm{U}} = \begin{pmatrix} 1.00 & 0.04 \\ 0.04 & 1.75 \end{pmatrix},$$
$$J_{\mathrm{L}} = \begin{pmatrix} 0.15 & 0 \\ 0 & -0.10 \end{pmatrix}, \qquad J_{\mathrm{U}} = \begin{pmatrix} 0.05 & 0 \\ 0 & -0.04 \end{pmatrix}, \qquad V = \begin{pmatrix} 0.12 & 0 \\ 0 & 0.10 \end{pmatrix}. \tag{34}$$

The Bloch matrix is

$$K(q;\rho) = \begin{pmatrix} D_{\mathrm{L}} + \rho(zJ_{\mathrm{L}} + z^{-1}J_{\mathrm{L}}^\dagger) & \rho z V \\ \rho z^{-1}V^\dagger & D_{\mathrm{U}} + \rho(zJ_{\mathrm{U}} + z^{-1}J_{\mathrm{U}}^\dagger) \end{pmatrix}, \qquad z = e^{2\pi i q}. \tag{35}$$

Thus $\rho = 0$ is the flat onsite endpoint and $\rho = 1$ is the fully dispersive parent; the five-scale surface uses $\rho \in \{0, 0.25, 0.5, 0.75, 1\}$. The parent has exact range one for $\rho > 0$. Its intersector coupling makes the lower two-dimensional spectral projector momentum dependent even though the parent itself is finite range. E_{ref} is only a normalization scale.

At every sampled momentum, the four Hermitian eigenpairs are sorted and the two lowest states are retained. The second-to-third band gap is monitored so that the retained rank and band identity are not changed silently. Raw eigensolver vectors are not used as a common coordinate system because their phases and internal rank-two rotations are independent at each point.

4.2 Parallel-transport frame and periodic sewing

Let $Y_i \in \mathbb{C}^{4\times 2}$ contain the raw retained eigenvectors at q_i, and let F_i be the accepted transported frame. Starting with $F_0 = Y_0$, form

$$M_i = F_i^\dagger Y_{i+1} = L_i \Sigma_i R_i^\dagger, \qquad F_{i+1} = Y_{i+1} R_i L_i^\dagger. \tag{36}$$

The rotation minimizes $\|Y_{i+1}Q - F_i\|_F$ over unitary Q. Both singular values of M_i must pass the declared rank gate; a rank-deficient overlap is rejected instead of being completed arbitrarily. Orthonormality and retained-projector reconstruction are checked at every node.

Transport around the periodic mesh generally leaves a sewing mismatch. The duplicate endpoint is transported once more, the unitary polar part of $S = F_0^\dagger F_N$ is diagonalized as $S = Z\,\mathrm{diag}(e^{i\gamma_1}, e^{i\gamma_2})Z^\dagger$, and the mismatch is distributed through

$$W_i = Z\,\mathrm{diag}(e^{-i(i/N)\gamma_1}, e^{-i(i/N)\gamma_2})Z^\dagger, \qquad F_i \leftarrow F_i W_i. \tag{37}$$

The corrected endpoint must equal the start within tolerance. The retained represented Hamiltonian is

$$H(q_i) = F_i^\dagger K(q_i) F_i. \tag{38}$$

The attack $U_a(q) = R[a\sin(2\pi q)]$ then gives $A_a(q) = U_a(q)^\dagger H(q) U_a(q)$ for $0 \le a \le \pi/4$. Complete spectra are equal by construction, while the comparison admits only one subsequent global real rotation.

4.3 Finite Fourier representation and range decision

For the centered half-open construction nodes $q_i = -\frac{1}{2} + i/N$, $i = 0, \ldots, N-1$, with uniform weights $\alpha_i = 1/N$ and transport anchor $q_0 = -1/2$,

$$T_R^{(N)} = \frac{1}{N}\sum_{i=0}^{N-1} e^{-2\pi i q_i R} H(q_i), \qquad H_N(q) = \sum_R e^{2\pi i q R} T_R^{(N)}. \tag{39}$$

Construction-node reconstruction, cyclic Hermiticity, and Parseval equality are checked. For even N, the centered Nyquist representative $-N/2$ is self-conjugate only in the cyclic convention. It is removed before ordinary-lattice truncation, and its power is reported as a conversion channel. The ordinary inventory is $\{-N/2+1, \ldots, N/2-1\}$ and must satisfy $T_{-R} = T_R^\dagger$.

For each integer cutoff R_c, the model

$$H^{\mathrm{ord}}_{N,R_c}(q) = \sum_{|R|\le R_c} e^{2\pi i q R} T_R^{(N)} \tag{40}$$

is evaluated on the shifted inventory

$$q_j^{\mathrm{eval}} = -\frac{1}{2} + \frac{j + 1/(N+1)}{N_{\mathrm{eval}}}, \qquad j = 0, \ldots, N_{\mathrm{eval}} - 1, \tag{41}$$

which is disjoint from the construction nodes. The headline endpoint uses $N = 512$ and $N_{\mathrm{eval}} = 1025$. Matrix maximum/RMS and ordered-band maximum/RMS errors are retained separately

for interpolation, deliberate truncation, and total finite-model consequences. For a band target τ, the first-passing range is

$$\widehat{R}^{\star}_{c,\mathrm{band}}(\tau) = \min\left\{ R_c : \max_{j,n} \frac{|\epsilon_n(H^{\mathrm{ord}}_{N,R_c}(q_j^{\mathrm{eval}})) - \epsilon_n(H(q_j^{\mathrm{eval}}))|}{E_{\mathrm{ref}}} \le \tau \right\}. \tag{42}$$

The Hamiltonian decision maps use the analogous maximum Frobenius error divided by E_{ref}. RMS channels are reported diagnostics but do not select the quoted cutoffs. The search is exact only on its bounded inventory; it is not a continuum localization length.

The flat analytic family is evaluated directly from the Bessel expressions. The non-flat parent uses directly sampled attack amplitudes rather than an interpolated continuum claim: 65 amplitudes for momentum-refinement diagnostics and 33 amplitudes for the parent/truncation surfaces. Momentum resolutions from 32 through 512 are retained, and the headline endpoint uses $N = 512$ with cutoffs $R_c = 0, \ldots, 16$.

4.4 Synthetic numerical results

For every ρ, $\kappa(\rho)$ is recomputed from the transported rank-two field using Eq. (19). The plotted symbol $\varepsilon_{\min}$ denotes the minimum globally aligned operator loss over the declared shift, scale, and real-rotation family; $P^H_{>0}$ and $P^A_{>0}$ denote the per-state non-onsite Fourier powers of the reference and attacked fields, respectively. Figure range labels use the same cutoff symbol R_c as Eq. (40).

At the flat endpoint $a = \pi/4$, the normalized minimum operator and exact-spectrum joint thresholds are approximately 0.8815978972 and 1.0276174232. The induced non-onsite power is approximately 0.7772149, consistent with the squared minimum threshold. The finite transform is roundoff-limited over the tested $N = 32, \ldots, 512$ sequence.

For the dispersive parent at $N = 512$, the sampled external gap is approximately $1.0566461 E_{\mathrm{ref}}$ and the minimum transport-overlap singular value is 0.9999997323. Complete bands agree to about $1.45 \times 10^{-15} E_{\mathrm{ref}}$, while the comparison non-onsite power rises from 0.0323839143 to 0.1761532501 over the attack.

The central finite-range result is measured at equal sampled accuracy. At a band-error target of $10^{-8} E_{\mathrm{ref}}$, the first passing cutoffs are

$$\widehat{R}^{\star}_{c,\mathrm{ref}} = 4, \qquad \widehat{R}^{\star}_{c,\mathrm{comp}} = 10. \tag{43}$$

The band-equivalent comparison therefore requires six additional lattice steps on the declared cutoff inventory. At targets 10^{-6}, 10^{-8}, 10^{-10}, and $10^{-12} E_{\mathrm{ref}}$, the endpoint reference/comparison pairs are $(3, 8)$, $(4, 10)$, $(6, 12)$, and $(7, 14)$.

The active rank-two calculation has a passing direct-v2 execution with independent byte-level reconstruction for its declared final-mesh parent family, scale-one truncation slice, and refinement channels. Its internal "unsealed" label means that the package has not completed the project's final immutable archival lifecycle; it is not a numerical failure. The package does not authenticate every five-scale development surface used in the broad preview figures. The exact theorem is not weakened by that lifecycle limitation; the numerical publication headline remains development or partial evidence until a complete-scope retained package is separately verified and archived.

5 Silicon source and reduction methodology

5.1 Source construction and adequacy boundary

The material-facing source is the authenticated $C6$-B20-W21 rank-10 Wannier Hamiltonian constructed from scalar-relativistic, non-spin-polarized PBE silicon calculations using Quantum ESPRESSO 7.5 and Wannier90 3.1.0 [20, 21, 22, 4]. The notation records a Gamma-origin 6^3 construction mesh, 20 parent bands, and a rank-10 retained space. Spinors are absent. The analysis therefore concerns a scalar-relativistic orbital Hamiltonian, not an SOC spinor Hamiltonian.

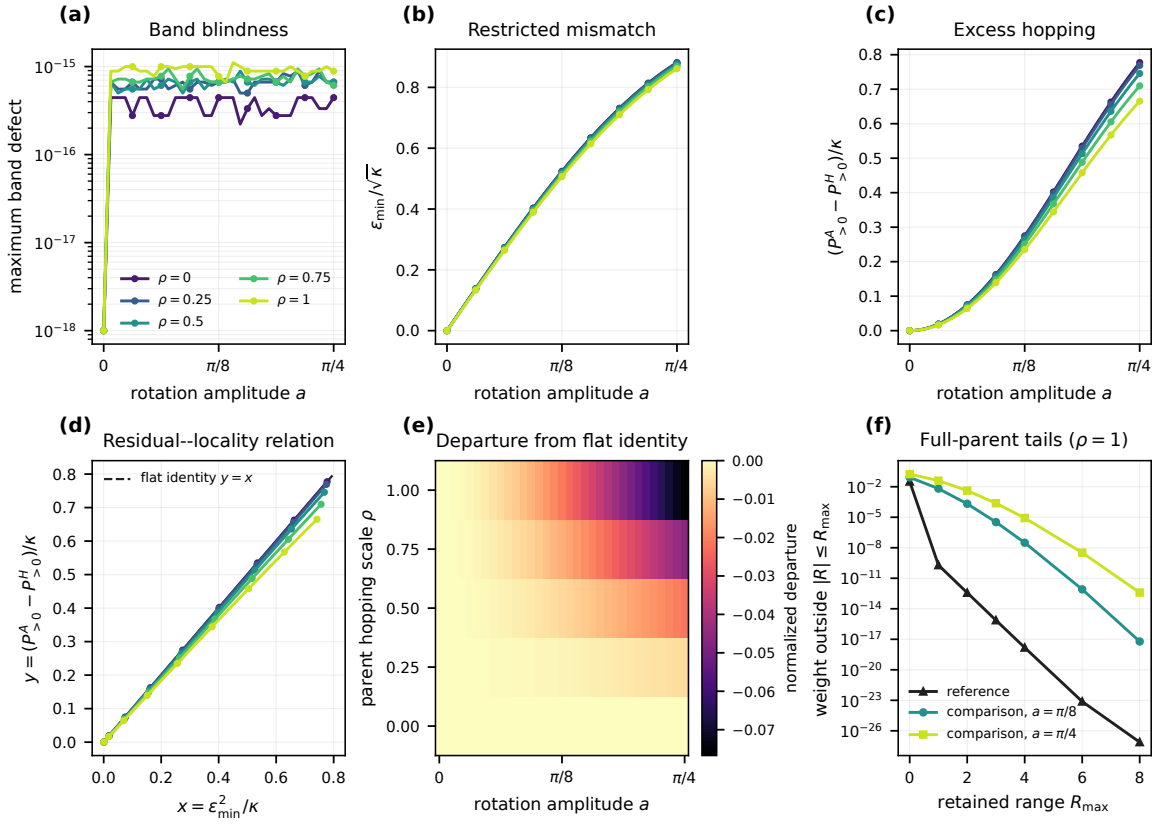


Figure 1: Controlled rank-two mechanism from exact band equivalence to representation-dependent hopping locality. The complete fields remain pointwise unitary equivalents; locality differences become finite-range consequences only after a support rule and accuracy criterion are imposed.

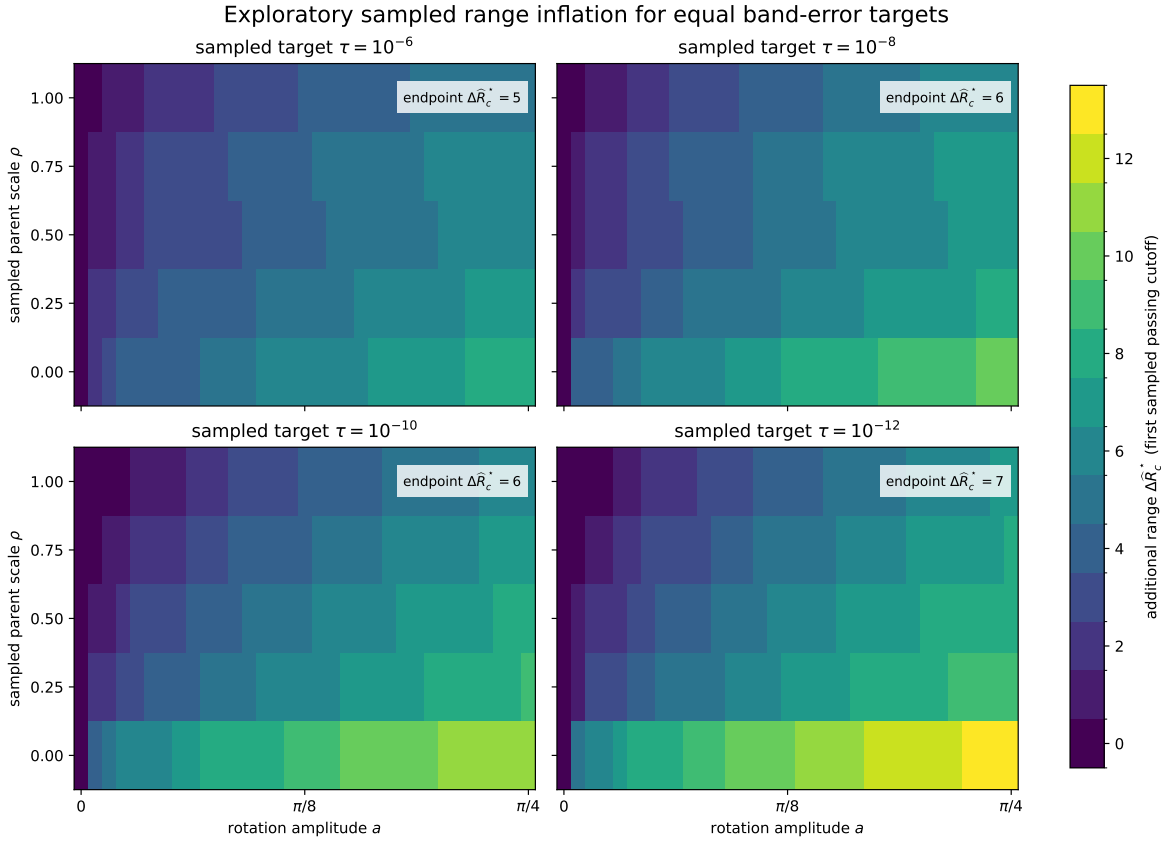


Figure 2: Sampled fixed-accuracy range inflation across parent scale, attack amplitude, and error target. The first-passing search is exact only over the declared integer cap $R_c = 0, \ldots, 16$; failure within the cap is not a universal no-solution result.

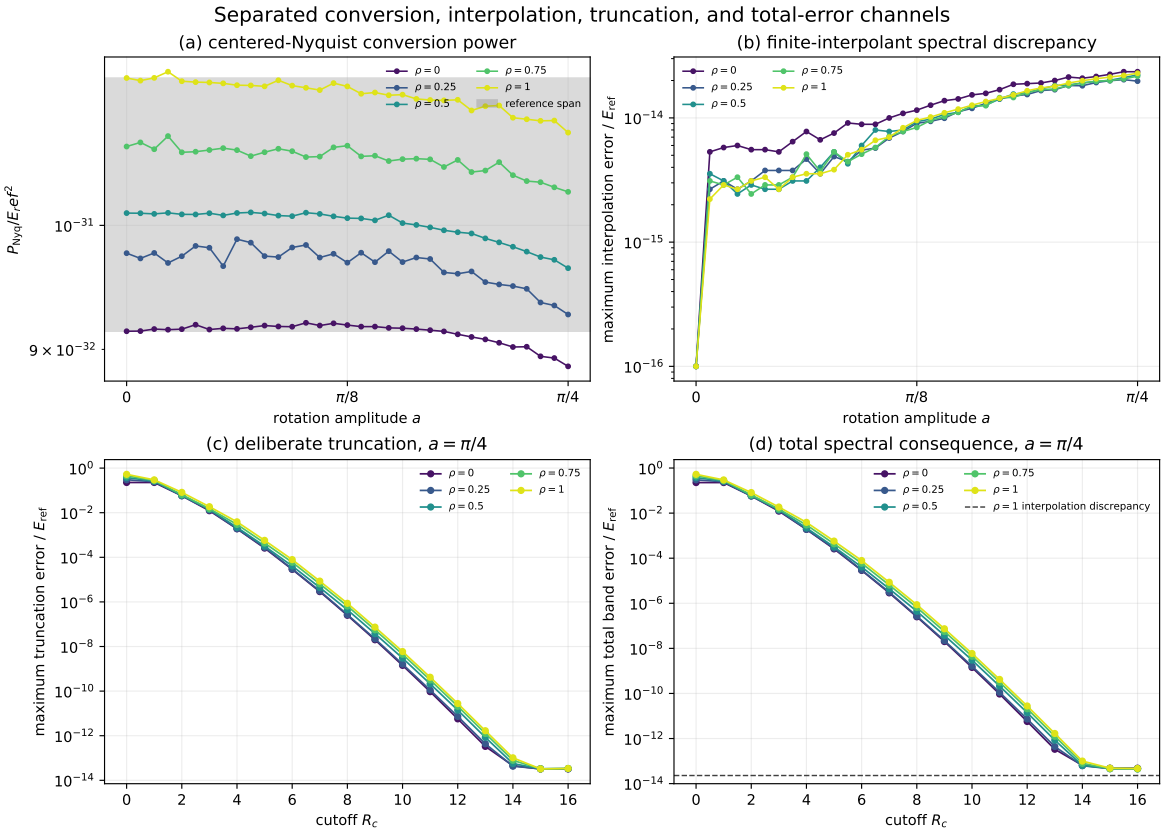


Figure 3: Separated finite-representation error channels in the rank-two calculation. Cyclic conversion, interpolation, deliberate truncation, and total consequences are displayed independently and are not treated as additive spectral errors.

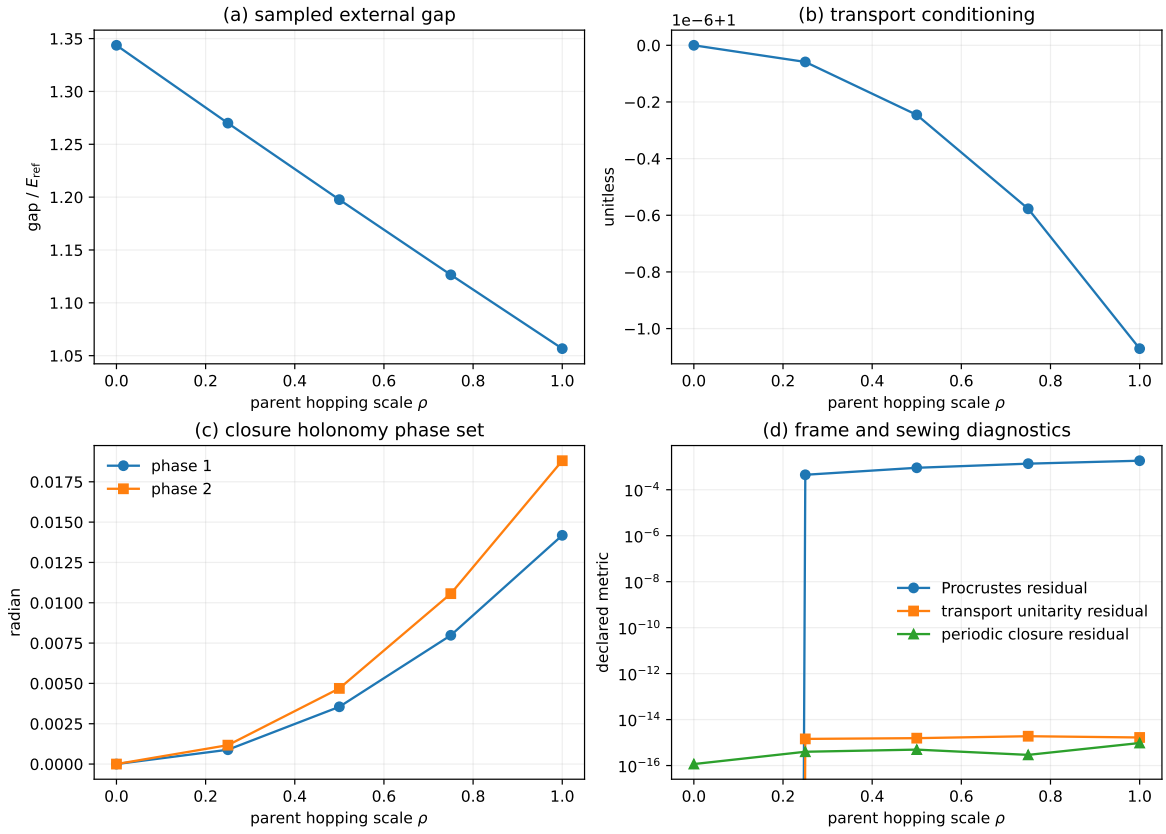


Figure 4: Retained-frame transport and sewing diagnostics for the four-state parent. The external gap, neighboring-frame overlap, orthonormality, projector reconstruction, and periodic closure are gates on the represented rank-two field.

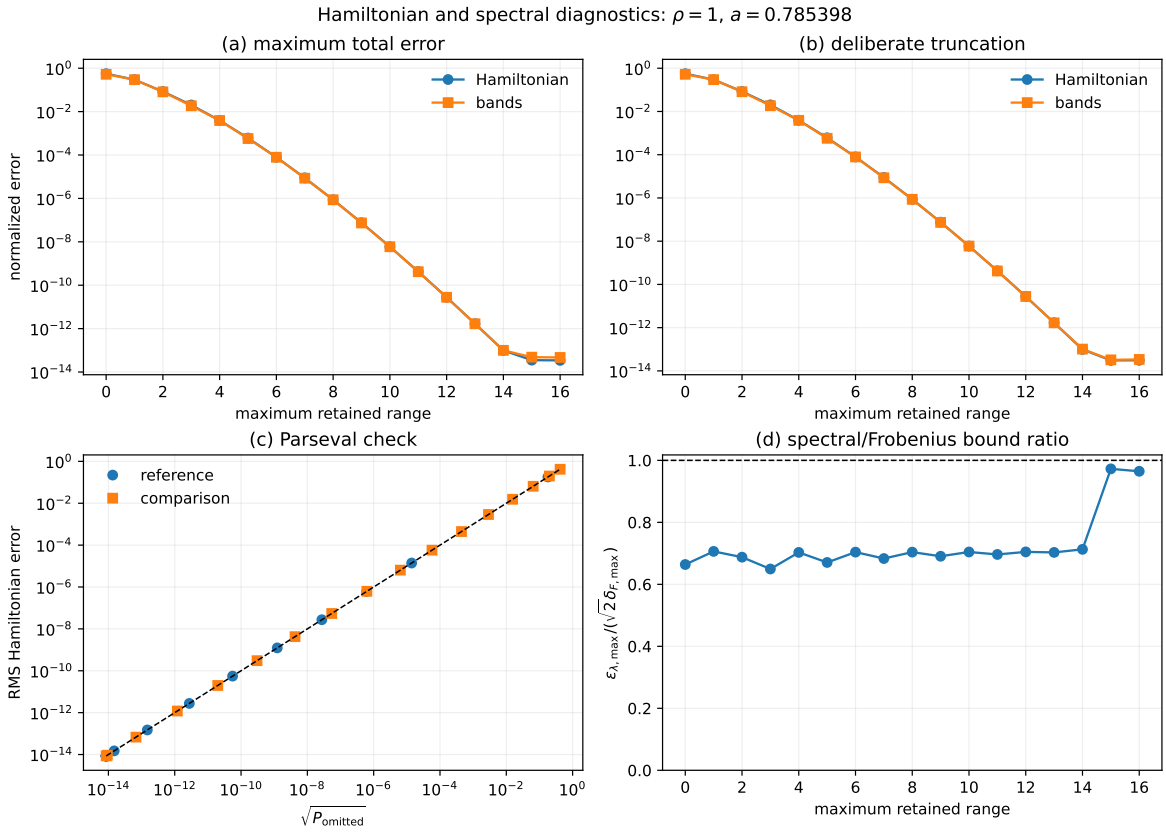


Figure 5: Hamiltonian and complete-spectrum diagnostics in the controlled calculation. The figure distinguishes matrix-field change from eigenvalue invariance before any finite-range approximation is imposed.

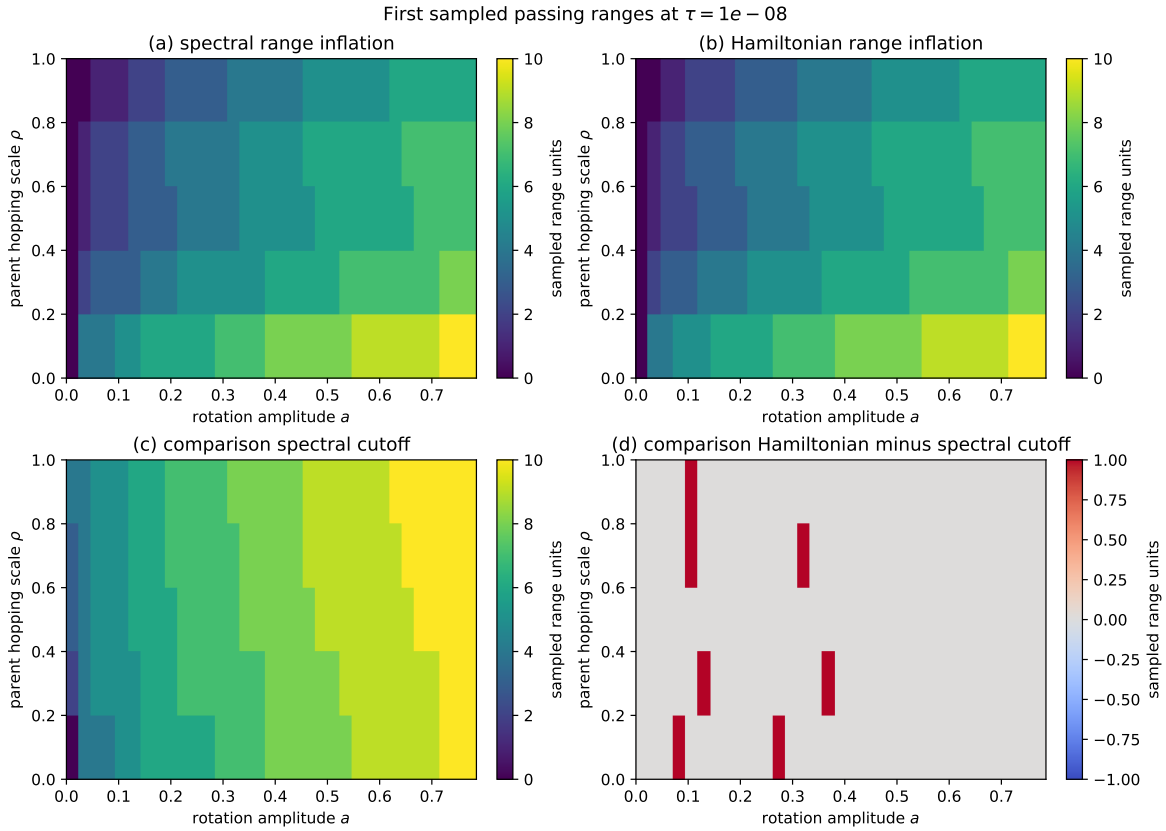


Figure 6: Sampled first-passing range decisions over the declared attack and parent-scale grid. Each cell reports a bounded integer search result, not a continuum optimum or a universal localization length.

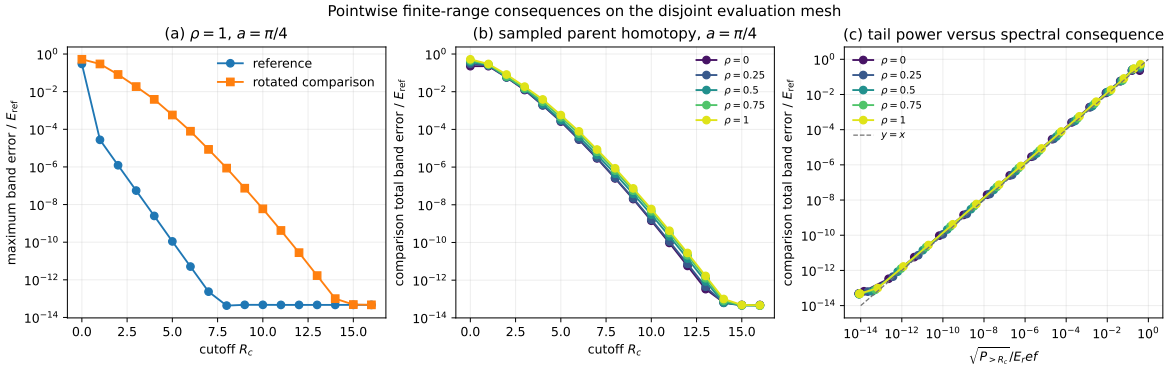


Figure 7: Band consequences of finite-range truncation in the controlled family. Reference and attacked representations are assessed at identical support rules and evaluation inventories.

The primitive cell contains Si atoms at fractional coordinates $(0,0,0)$ and $(1/4,1/4,1/4)$. The relaxed right-handed FCC primitive basis, in angstrom, is

$$A = \begin{pmatrix} 2.73459104682 & 0 & 2.73459104682 \\ 2.73459104682 & 2.73459104682 & 0 \\ 0 & 2.73459104682 & 2.73459104682 \end{pmatrix}, \tag{44}$$

where the primitive vectors are the columns of A, and the corresponding conventional cubic lattice parameter is $a = 5.469182093640207$ Å. The reciprocal-basis columns are

$$B = \begin{pmatrix} 1.14883454228 & -1.14883454228 & 1.14883454228 \\ 1.14883454228 & 1.14883454228 & -1.14883454228 \\ -1.14883454228 & 1.14883454228 & 1.14883454228 \end{pmatrix} \text{ Å}^{-1}. \tag{45}$$

The cell was obtained from an authorized FCC-native `vc-relax` with `ibrav=2`, initial $A = 5.43$ Å, and scalar-cell relaxation. Three BFGS steps and four structural SCF cycles reached a reported terminal pressure of 0.01 kbar and a final-SCF pressure of 0.09 kbar under a 0.5 kbar declared tolerance. These observations establish the retained geometry occurrence; they do not constitute a general structural uncertainty analysis.

The accepted parent SCF used the SSSP 1.3.0 PBE-efficiency silicon ultrasoft pseudopotential [23], a Gamma-centered 10^3 mesh, and wavefunction/density cutoffs of 42/336 Ry. The retained occurrence reported electronic convergence in five iterations. The Wannier parent NSCF used the same accepted structure and cutoffs, 216 ordered Gamma-origin 6^3 points, 20 bands, and Davidson diagonalization. Its eigenvalue range was $[-5.83413, 30.27201]$ eV. These calculations were performed and authorized in their own retained campaign records; they were not rerun for the present post-hoc study.

Wannier90 retained ten functions from the 20-band inventory. The input used `spinors=.false.`, outer disentanglement window $[-6, 21]$ eV, frozen window $[-5.85, 8]$ eV, and projections `Si:sp3` together with one additional radial s-like projection per Si specified by `Si:l=0:r=2`. Both disentanglement and Wannier localization had 40000-step ceilings and 2×10^{-9} convergence gates. The retained run converged at disentanglement iteration 1517 and Wannier iteration 378, with final total spread 21.022751219 Å^2. This is native optimizer convergence, not evidence of a global spread minimum.

The native HR file contains 279 Wannier90 representatives. The authenticated `wsvec` record supplies, for each native representative $\mathbf{R}$ and matrix entry (m,n), a nonempty set of minimum-distance integer shifts $\mathcal{S}_{\mathbf{R}mn}$. The effective ordinary coefficient is constructed entrywise as

$$[T_{\mathrm{W}}(\mathbf{R}')]_{mn} = \sum_{\substack{\mathbf{R},\mathbf{s}\in\mathcal{S}_{\mathbf{R}mn} \\ \mathbf{R}+\mathbf{s}=\mathbf{R}'}} \frac{[H_{\mathrm{HR}}(\mathbf{R})]_{mn}}{d_{\mathbf{R}}\,|\mathcal{S}_{\mathbf{R}mn}|}, \qquad H_{\mathrm{W}}(\boldsymbol{\kappa}) = \sum_{\mathbf{R}'} e^{2\pi i \boldsymbol{\kappa}\cdot\mathbf{m}(\mathbf{R}')} T_{\mathrm{W}}(\mathbf{R}'). \tag{46}$$

Thus a boundary HR entry is divided equally among its declared minimum-distance replicas as well as by the native block degeneracy. Collecting equal resulting translations, sorting the integer triples, and retaining zero blocks needed by the elementwise map produces 383 effective ordinary representatives from the 279 native HR representatives. This is the retained `use_ws_distance` route; merely dividing each native block by $d_{\mathbf{R}}$ would still leave 279 blocks and is not the reconstruction used here. Pair closure, Hermiticity, construction-mesh reconstruction, reciprocal periodicity, and finite spectra were checked before the source was admitted to reduction analysis.

Structural eligibility does not imply material adequacy. In the canonical corrected current-source comparison against the revealed 408-point DFT parent, the source indirect gap is 0.6892447 eV and the parent gap is 0.6112467 eV, a 12.7605% relative difference against the prospectively declared 5% target. All six valley positions pass their 0.034465 Å^{-1} displacement target, with displacements from approximately 0.01906 to 0.02134 Å^{-1}, but all six longitudinal masses and eleven of twelve transverse masses fail the 8% target. The non-gating injective

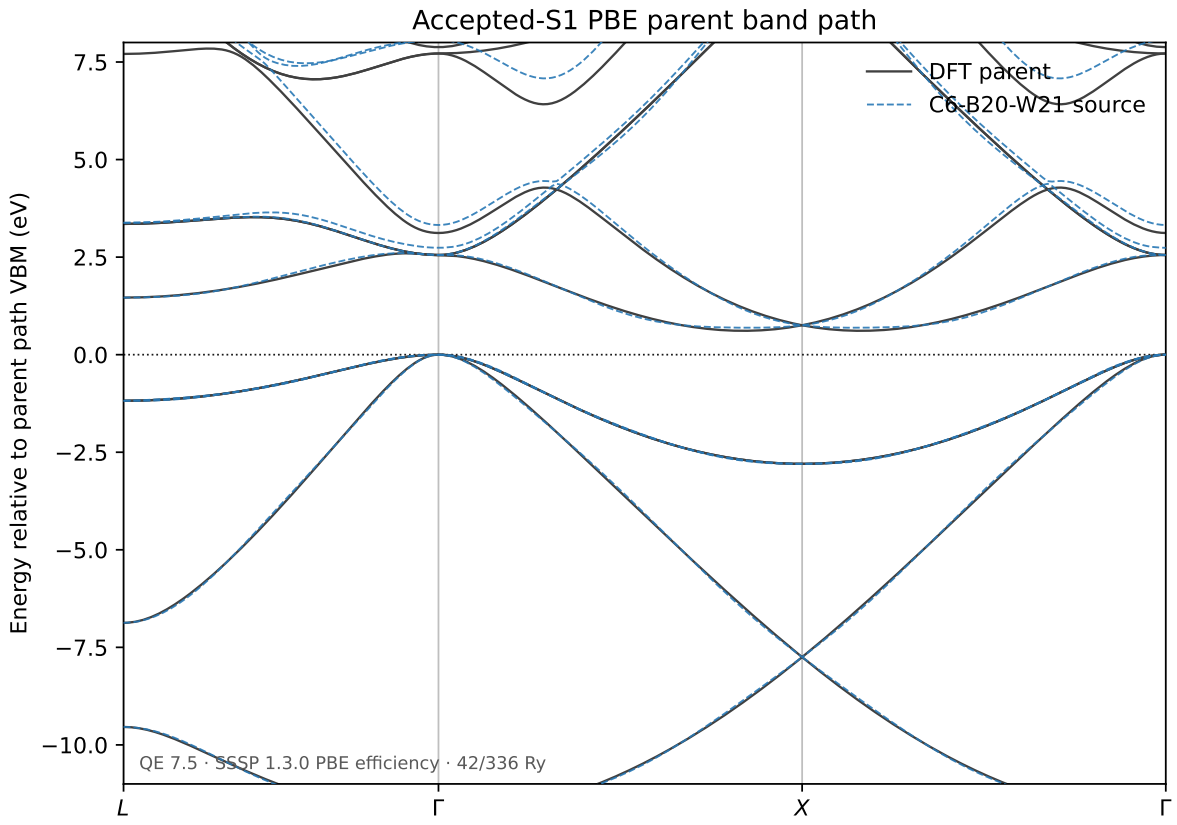


Figure 8: Accepted-S1 parent DFT band path and full Wannier source. The solid curves are the lowest 12 of 20 scalar-relativistic PBE parent bands; dashed curves are the ten bands of the authenticated $C6$–$B20$–$W21$ source. The horizontal path contains 241 ordered points. The common display zero is the parent path VBM, and no fitted energy alignment is used.

408-point spectral residual is 0.246196 eV RMS and 0.737286 eV maximum. These failures bound every result below: source-relative preservation is not DFT validation.

An earlier parent-comparison record reported a 0.5226105 eV candidate gap. The retained source identities resolve the apparent discrepancy: that value belongs to a distinct lower-energy-frozen $4^3/12$-band Wannier candidate whose eligible HR hash is `2139eb7f...`, not to the present $C6$–$B20$–$W21$ candidate whose HR hash is `24a411c8...`. The two records share the same revealed DFT parent gap but do not share a candidate Hamiltonian. The present article therefore uses 0.6892447 eV only for the authenticated $C6$–$B20$–$W21$ source and does not combine the candidate lineages.

No frozen 402-point confirmation values are used. No Quantum ESPRESSO, interface, or Wannier90 process was invoked for the gauge, shell, fidelity, or band-edge analyses. Every subsequent Hamiltonian is reconstructed from the same authenticated HR and Wigner–Seitz data in the same rank-10 Wannier basis and native energy reference.

For band-structure context, one separately authorized fixed-density QE 7.5 calculation was performed after the referee revision. It consumed an isolated copy of the accepted K10 SCF state and evaluated 20 parent Kohn–Sham bands at 241 ordered points on L–Γ–X–Γ. Figure 8 displays the lowest 12 parent bands together with the complete rank-10 Wannier source. Both curves retain their native common energy reference; the plot subtracts the parent path VBM and applies no fitted source shift. The sampled path gaps are 0.6114 eV for the parent and 0.69046 eV for the source. These rounded path diagnostics visualize, but do not replace, the separately optimized indirect-gap records.

5.2 Higher-rank gauge attack

The internal attack is frozen independently of the silicon Hamiltonian. It is a real orthogonal—and therefore complex-unitary—rank-10, depth-2 alternating nearest-neighbor Givens circuit in the authenticated Wannier coordinate order. The indices below refer only to that retained coordinate order; no atomic-orbital identity is assigned to an index.

$$G(\mathbf{k}) = G_1(\mathbf{k})G_2(\mathbf{k}), \qquad A(\mathbf{k}) = G(\mathbf{k})^\dagger H_{\mathrm{W}}(\mathbf{k})G(\mathbf{k}). \tag{47}$$

The layer index is one-based: $\ell \in \{1, 2\}$. Layer 1 rotates pairs $(0, 1), (2, 3), (4, 5), (6, 7), (8, 9)$; layer 2 rotates $(1, 2), (3, 4), (5, 6), (7, 8)$. For every active pair $(p, p+1)$ the embedded real rotation block is

$$\begin{pmatrix} \cos\theta_{\ell p} & -\sin\theta_{\ell p} \\ \sin\theta_{\ell p} & \cos\theta_{\ell p} \end{pmatrix}. \tag{48}$$

Let the harmonic bonds be $\mathbf{b}_1 = (1, 0, 0)$, $\mathbf{b}_2 = (0, 1, 0)$, and $\mathbf{b}_3 = (0, 0, 1)$. The shared layer scalar is

$$g_\ell(\mathbf{k}) = \frac{1}{3}\sum_{a=1}^{3}\left[\frac{3}{4}\sin(2\pi\mathbf{k}\cdot\mathbf{b}_a) + \frac{1}{4}(-1)^\ell \sin(4\pi\mathbf{k}\cdot\mathbf{b}_a)\right]. \tag{49}$$

For a pair whose right index is p, the rotation angle is

$$\theta_{\ell p}(\mathbf{k}) = \frac{\pi}{8}(-1)^{\ell+p}\frac{1 + 0.15(2p-10)/9}{1.15}\, g_\ell(\mathbf{k}). \tag{50}$$

This construction is periodic, has conservative angle bound $\pi/8$, and supplies its exact inverse $G(\mathbf{k})^\dagger$. It was selected as a synthetic middle-amplitude control before inspecting the silicon support results; changing a bond, depth, normalization, or coefficient defines a different attack. It is a constructed stress test demonstrating one allowed redistribution of locality. It is not claimed to be representative of typical Wannier90 gauge variation or to maximize range inflation.

Here $\boldsymbol{\kappa}$ denotes dimensionless reduced reciprocal coordinates and the $\mathbf{b}_a$ are integer harmonic vectors, so every trigonometric argument is $2\pi\boldsymbol{\kappa}\cdot\mathbf{b}_a$. Cartesian wavevectors are denoted explicitly by $\mathbf{k}_{\mathrm{cart}} = B\boldsymbol{\kappa}$ and have units of Å^{-1}.

The source construction coordinates are the native Gamma-origin points $0, 1/6, \ldots, 5/6$ on each reduced reciprocal axis. For the centered transform they are wrapped and permuted to

$$\mathcal{K}_6 = \{-1/2, -1/3, -1/6, 0, 1/6, 1/3\}^{\otimes 3}, \tag{51}$$

with zero construction offset. The direct native HR/Wigner–Seitz route is evaluated first; the generic field adapter does not reconstruct the source from raw DFT wavefunctions. The distinct evaluation inventory is the ordered 408-point development geometry with uniform weights $w_j = 1/408$. It is a candidate-valley-centered derivative inventory rather than a tensor grid: six candidate centers, 72 capture-axis offsets at 0.02 and 0.04 Å^{-1}, 108 axial Hessian offsets at 0.002, 0.005, and 0.01 Å^{-1}, 216 mixed-Hessian offsets at the same three steps, and six signed-axis symmetry anchors at the mean candidate radius. Duplicate coordinates are merged by ordered point identity, leaving 408 unique points and zero overlap with the earlier Gamma-origin 4^3 construction mesh. The point list is represented in the accepted primitive reciprocal basis and wrapped to $[0, 1)^3$. The frozen 402-point confirmation geometry was generated separately, is construction-disjoint, and its values play no role.

Four alignment channels are evaluated. Identity uses the common reference coordinates. Global-unitary alignment stacks the admitted rank-10 construction frames and selects one right unitary by the aggregate polar/Procrustes problem in Eq. (6); no nonlinear optimization over a parameterization of $U(10)$ is performed. Pointwise polar alignment obtains one unitary per coordinate from the admitted frame overlaps. Known-inverse alignment applies the exact inverse

action of Eq. (47). The frozen rank cutoffs use the additive rule with absolute threshold 10^{-12} and relative threshold 10^{-10}; the same absolute and relative values are used for transport-rank checks. A deficient overlap fails rather than receiving an implicit null-space completion. Pointwise polar and known inverse are oracle controls, not candidate discovery procedures.

For a centered cyclic displacement $\mathbf{m}$, the normalized finite transform is

$$T_{\mathbf{m}}^{(6)} = \frac{1}{216} \sum_{\mathbf{k} \in \mathcal{K}_6} e^{-2\pi i \mathbf{k}\cdot\mathbf{m}} H(\mathbf{k}), \qquad H^{(6)}(\mathbf{k}) = \sum_{\mathbf{m}} e^{2\pi i \mathbf{k}\cdot\mathbf{m}} T_{\mathbf{m}}^{(6)}. \tag{52}$$

Construction-point reconstruction, block Hermiticity, and Parseval consistency are checked. Because an even finite grid contains Nyquist representatives whose ordinary integer conjugate is not represented independently, cyclic blocks are then converted explicitly to an inversion-complete ordinary inventory. Conversion-removed power is recorded separately. For this size-six route the conversion is an exact selection rule: enumerate the 216 centered cyclic modes on $\{-3,-2,-1,0,1,2\}^{\otimes 3}$, remove every mode having at least one $-3 = -N/2$ component, and map every remaining mode identically into $\{-2,-1,0,1,2\}^{\otimes 3}$. This removes 91 negative-Nyquist-boundary records and retains 125 ordinary blocks before support truncation. A machine-readable ledger now records every cyclic index, conjugate representative, retain/remove decision, ordinary index, FCC metric label, physical radius, and retained per-state block powers. An independent integer-inventory audit reproduces the 125-mode ordering exactly and the physical radii to 3.56×10^{-15} Å. It does not reconstruct the 216 cyclic coefficient matrices or removed-mode power, which remain part of the broader Fourier reconstruction blocker.

Complete supports are defined by the FCC metric. For $\mathbf{m} = (n_1, n_2, n_3)$,

$$q(\mathbf{m}) = 2(n_1^2 + n_2^2 + n_3^2 + n_1 n_2 + n_1 n_3 + n_2 n_3), \qquad r(\mathbf{m}) = \frac{a}{2}\sqrt{q(\mathbf{m})}. \tag{53}$$

The formula follows directly from $\mathbf{R} = A\mathbf{m}$. For the FCC primitive basis in Eq. (44),

$$\|A\mathbf{m}\|^2 = \frac{a^2}{2}(n_1^2 + n_2^2 + n_3^2 + n_1 n_2 + n_1 n_3 + n_2 n_3) = \frac{a^2}{4} q(\mathbf{m}), \tag{54}$$

which is equivalent to Eq. (53). The attained labels are ordered up to $q = 48$; every support contains all blocks at or below one attained label and is inversion complete. These are distances between lattice-block representatives. They are not orbital-center or individual matrix-entry distances. This choice preserves an exact blockwise Fourier/Parseval accounting and avoids assigning atomic-orbital semantics to the authenticated Wannier basis, but it is deliberately coarser than entry-magnitude pruning or orbital-pair minimum-distance cutoffs such as `hr_cutoff`, `dist_cutoff`, and minimum-distance replica selection [4, 24]. The reported support fractions must therefore be read as properties of this whole-block lattice-displacement family, not as lower bounds for all possible tight-binding parameterizations.

For each representation and support, the producer retains maximum and RMS matrix Frobenius errors, maximum and RMS sorted-spectrum errors, omitted block power, physical radius, displacement inventory, and block count. The matrix RMS is normalized by $\sqrt{10} \times 1$ eV and the sorted-spectrum RMS by the same per-state scale. The calculation distinguishes:

1. *direct-parent error*: the complete Wannier source relative to revealed DFT eigenvalues, without defining a DFT operator subtraction in the Wannier basis;

2. *exact gauge equivalence*: the complete source and attacked fields before finite Fourier representation;

3. *conversion*: the 91 centered cyclic blocks touching the negative Nyquist boundary that are removed before the remaining 125 modes map identically;

4. *interpolation*: the full 125-block ordinary interpolant relative to the direct native represented target on the 408 points;

Table 1: Complete native Wigner–Seitz shell hierarchy. The last column is cumulative support.

Shell	Radius (Å)	New representatives	Cumulative representatives
H-shell-00	0.000000000	1	1
H-shell-01	3.867295746	12	13
H-shell-02	5.469182094	6	19
H-shell-03	6.698352720	24	43
H-shell-04	7.734591492	12	55
H-shell-05	8.647536177	24	79
H-shell-06	9.472901262	8	87
H-shell-07	10.231902790	48	135
H-shell-08	10.938364187	6	141
H-shell-09	11.601887238	36	177
H-shell-10	12.229462943	24	201
H-shell-11	12.826368943	24	225
H-shell-12	13.396705440	24	249
H-shell-13	13.943733109	64	313
H-shell-14	14.977972019	40	353
H-shell-15	15.945268846	12	365
H-shell-16	16.407546281	6	371
H-shell-17	16.857151341	12	383

5. *deliberate truncation*: a support-limited ordinary model relative to its full ordinary interpolant; and

6. *represented total*: the final support-limited model relative to the direct native represented source.

Separately maximized or RMS channels are never added as if they were signed scalar errors. Complete support curves are compared by observed discrete non-dominance and by "equal-attained-error" queries: for an error actually attained by one discrete support, the minimum support of the other representation attaining no larger error is reported. No interpolation between support counts and no post-hoc acceptance threshold is used. Native Wigner–Seitz shells remain a separate fixed-gauge route.

5.3 Native shell hierarchy and literal reduction

The effective native representatives are grouped by Cartesian radius after rounding the radius to nine decimal places. This produces 18 nested, inversion-complete shells. Table 1 gives the complete family. H-shell-00 is the onsite control and H-shell-17 is the full 383-representative source. Literal reductions H-shell-00 through H-shell-16 retain the source coefficients unchanged inside one shell boundary and set every coefficient outside it to zero. Full support remains a zero-loss control and is excluded from proper-reduction feasibility.

For every literal candidate, $H_S(\mathbf{k})$ is reconstructed directly on the ordered 408-point inventory in the common Wannier basis. The losses are Eqs. (12)–(13); no DFT parent values are consumed. Maximum ordered-spectrum and per-state Frobenius errors are retained as additional coordinates. A threshold cell (t_1, t_2) is populated by the smallest proper support with $\tau_1 \leq t_1$ and $\tau_2 \leq t_2$. Because the cells are evaluated only after all candidates are known, the resulting surface is descriptive rather than a prospective selection policy.

5.4 Refitted complete-shell family and analytic gradients

The refitted family retains the same supports and Wannier basis but assigns one bounded real scale to each complete retained shell. Define the shell-resolved field

$$H_s(\mathbf{k}) = \sum_{\mathbf{R}\in s} e^{i\mathbf{k}\cdot\mathbf{R}} T_{\mathrm{W}}(\mathbf{R}). \tag{55}$$

For support $S_p = \{0, \ldots, p\}$,

$$H^{(p)}_{\boldsymbol{\alpha}}(\mathbf{k}) = \sum_{s=0}^{p} \alpha_s H_s(\mathbf{k}), \qquad 0 \le \alpha_s \le 2. \tag{56}$$

The same α_s applies to every representative in shell s, including both $\mathbf{R}$ and $-\mathbf{R}$. Blocks outside S_p remain zero. Sign reversal and more-than-doubling are excluded. No matrix entry is fitted independently, and no atomic-orbital or Slater–Koster semantics are introduced.

Let $D_j = H_{\boldsymbol{\alpha}}(\mathbf{k}_j) - H_{\mathrm{W}}(\mathbf{k}_j)$ and let $\delta\epsilon_{jn}$ be the ordered eigenvalue difference. With $N_k = 408$, rank $N = 10$, and $E_0 = 1$ eV,

$$\tau_2^2 = \frac{1}{N_k N E_0^2} \sum_j \|D_j\|_F^2 \,, \tag{57}$$

$$\tau_1^2 = \frac{1}{N_k N E_0^2} \sum_{j,n} \delta\epsilon_{jn}^2, \tag{58}$$

$$\frac{\partial \tau_2^2}{\partial \alpha_s} = \frac{2}{N_k N E_0^2} \sum_j \mathrm{Re}\,\mathrm{tr}[D_j^\dagger H_s(\mathbf{k}_j)], \tag{59}$$

$$\frac{\partial \tau_1^2}{\partial \alpha_s} = \frac{2}{N_k N E_0^2} \sum_{j,n} \delta\epsilon_{jn} \langle v_{jn}| H_s(\mathbf{k}_j) |v_{jn}\rangle. \tag{60}$$

Equation (60) is the local Hellmann–Feynman derivative for the computed ordered eigensystem. At an exact degeneracy, individual diagonal expectations depend on the eigensolver's basis inside the degenerate subspace; the invariant first-order splittings are instead the eigenvalues of the projected perturbation block. The ordered spectral objective is consequently nonsmooth at crossings or degeneracies, and the retained expression supplies a local numerical subgradient away from unresolved crossings rather than a proof of global differentiability. This limitation is one reason the multistart results are reported only as constructive witnesses.

For each support and scalarization weight w, the optimized objective is

$$F_w(\boldsymbol{\alpha}) = w\tau_1^2 + (1-w)\tau_2^2. \tag{61}$$

The 17 weights are

$$w \in \{0, 0.001, 0.003, 0.01, 0.03, 0.1, 0.2, 0.35, 0.5, 0.65, 0.8, 0.9, 0.97, 0.99, 0.997, 0.999, 1\}. \tag{62}$$

Bounded L-BFGS-B uses at most 300 iterations, gradient tolerance 10^{-10}, and function tolerance 10^{-14}. Weights are swept in the ascending order displayed above. Three deterministic starts are used: the literal $\boldsymbol{\alpha} = \mathbf{1}$ candidate; an alternating vector beginning with the onsite scale $\alpha_0 = 0.75$, followed by $1.25, 0.75, \ldots$; and continuation from the best scalarized result at the preceding smaller weight. Scaling the complete onsite block also scales its trace and therefore couples centroid and internal onsite structure; no independent global energy-shift parameter is fitted. The analytic gradient was checked by finite differences; the maximum absolute discrepancy was 9.7361×10^{-10} and the maximum scaled discrepancy was 4.0729×10^{-11}. Only optimizer-success records enter the final witness surface. Among the 444 retained successful records, 50 have one coefficient active at the lower bound and 35 have one coefficient active at the upper bound; no retained record

has more than one active lower or upper coefficient. Bound activity occurs only on H-shell-02 (17 upper-bound records), H-shell-06 (50 lower-bound records), and H-shell-08 (18 upper-bound records). The bounds therefore shape parts of the observed witness set and remain part of the model definition. The 444 records are constructive witnesses; a missing witness is not a global infeasibility certificate. Within each post-hoc threshold cell the frozen record is chosen first by minimum representative count, then by the smallest maximum of the two attained-to-threshold ratios, then by smaller τ_1, and finally by smaller τ_2.

This is a shell-renormalized Wannier family, not a Slater–Koster model [25, 26]. It also imposes no crystallographic group-averaging constraint. In a two-sublattice crystal, a space-group operation can map one inter-orbital pair into a pair represented at a different Bravais translation; grouping or independently rescaling entire blocks only by $\|\mathbf{R}\|$ can therefore split a symmetry orbit across shell indices. The large moderate-radius Gamma splittings and X/L spreads may contain preventable broken-cancellation contributions in addition to finite-range error. Symmetry-adapted Wannier construction, post-hoc hopping symmetrization, and linear symmetry constraints during fitting are important untested baselines [27, 28, 29]. The present study does not determine how much support those alternatives would require. The separate Slater–Koster path stopped at a rank-9 common-space gate at Gamma and one additional construction point and produced no fitted parameter vector.

5.5 Frozen fidelity ladder

Three converged witnesses were selected from post-hoc threshold cells and frozen before broader evaluation: H-shell-05 with 79 representatives from cell $(0.1, 0.15)$, H-shell-10 with 201 representatives from $(0.02, 0.04)$, and H-shell-16 with 371 representatives from $(0.001, 0.002)$. Their exact shell-scale vectors, ordered from the onsite shell outward, are

$$\boldsymbol{\alpha}_{05} = (0.99939463, 1.02717894, 1.15464602, 0.96389478, 0.87025477, 0.84340786), \tag{63}$$

$$\begin{aligned}\boldsymbol{\alpha}_{10} = (&0.99931068, 1.00108654, 1.01987595, 1.04502720, 0.98170703, 0.92902579, 0.70419673,\\ &1.04750727, 0.57744398, 0.99055868, 1.15302212),\end{aligned} \tag{64}$$

$$\begin{aligned}\boldsymbol{\alpha}_{16} = (&1.00002718, 1.00000490, 1.00195671, 1.00018218, 0.99843025, 0.99840401, 0.99896498,\\ &0.99950810, 0.99768267, 1.00065532, 0.99738969, 0.99854754, 1.00013810, 1.00004865,\\ &1.01699983, 1.04338006, 1.00871625).\end{aligned} \tag{65}$$

All later calculations evaluate these coefficients without refitting.

Table 2 separates construction, fitting, selection, and evaluation inventories. In particular, the same 408 coordinates serve as the post-hoc shell-loss inventory, but their DFT parent eigenvalues are not used to fit shell scales or select the three frozen witnesses.

The ladder has three evaluation levels. First, exact normalized-Brillouin-zone operator RMS is computed from Eq. (9) using the unique effective native coefficients. Second, spectra and sampled operator residuals are evaluated on independent Gamma-origin tensor products

$$\mathcal{G}_N = \{(i/N, j/N, k/N) : i, j, k = 0, \ldots, N-1\}, \qquad N \in \{12, 20, 32\}. \tag{66}$$

The retained metrics are all-band RMS, maximum, and 95th-percentile absolute ordered spectral errors; lowest-conduction-band errors; and sampled per-state operator RMS. Mesh comparisons are made after the models are frozen and are therefore interpolation tests, not additional fitting data.

Third, six signed Δ-family sector minima are identified operationally. Each point of the Gamma-origin $[0, 1)^3$ grid is first mapped to its closest periodic image to Gamma in Cartesian reciprocal coordinates, and the resulting centered vector is partitioned by dominant signed Cartesian axis. For each of $\pm x$, $\pm y$, and $\pm z$, the eight lowest coarse points in the corresponding sector are retained as starts, together with the source valley coordinate. From each individual

Table 2: Data and reciprocal inventories by role. "Excluded" means unavailable to model construction, fitting, selection, and the reported development evaluation.

Inventory	Construction/fitting role	Selection role	Evaluation role
Relaxed cell, K10 SCF, C6-B20 NSCF	Defines the parent occurrence and Wannier inputs	None in the post-hoc reduction	Provenance and source-adequacy context
Gamma-origin 6^3 (216 points)	Wannier construction and finite-cyclic gauge transform	Attack frozen before silicon support outcomes	Construction reconstruction and Parseval checks
Ordered 408-point geometry	Shell loss and scalarized refit inventory; uniform weight 1/408	Post-hoc threshold cells identify frozen witnesses; no DFT parent values used	Gauge support curves and source-relative fitted-inventory losses
Gamma-origin 12^3, 20^3, 32^3	None	None; models already frozen	Dense spectra, sampled operator RMS, and mesh stability
32^3 signed-axis sectors plus local searches	None	None; models already frozen	Operational conduction sector minima and curvature diagnostics
Parent L–Γ–X–Γ path (241 points)	None; fixed-density parent visualization after model freezing	None	Contextual DFT/source band diagram; not a fitting or global-extremum inventory
Frozen 402-point confirmation set	Excluded	Excluded	Excluded; values remain unobserved

start, band 4 in zero-based ordering is minimized by L-BFGS-B over a Cartesian displacement box $[-0.12, 0.12]^3$ Å^{-1} centered on that start, with at most 240 iterations, function tolerance 10^{-14}, and gradient tolerance 10^{-10}. Periodic reduced coordinates are reconstructed using B^{-1}, and a candidate is retained only if its final dominant signed-axis label matches the requested valley identity. The lowest successful matching minimum is selected as the identified sector minimum; a bound-activity flag is retained. This bounded multistart procedure does not prove that the global minimum of the entire sector has been found. The eight-start sector search replaced a preliminary source-centered 0.08 Å^{-1} search that reached its bound for the H-shell-05 y valleys.

At each identified sector minimum $\mathbf{k}_v^*$, the primary Cartesian Hessian uses step $h = 0.002$ Å^{-1}. Diagonal and mixed entries are

$$K_{ii} = \frac{E(\mathbf{k}_v^* + h\mathbf{e}_i) - 2E(\mathbf{k}_v^*) + E(\mathbf{k}_v^* - h\mathbf{e}_i)}{h^2}, \tag{67}$$

$$K_{ij} = \frac{E(\mathbf{k}_v^* + h\mathbf{e}_i + h\mathbf{e}_j) - E(\mathbf{k}_v^* + h\mathbf{e}_i - h\mathbf{e}_j) - E(\mathbf{k}_v^* - h\mathbf{e}_i + h\mathbf{e}_j) + E(\mathbf{k}_v^* - h\mathbf{e}_i - h\mathbf{e}_j)}{4h^2}. \tag{68}$$

Diagonalization gives principal curvatures and axes. Principal masses in electron-mass units are $(\hbar^2/m_e)/\kappa_i$ after conversion from eV Å^2. Gap, valley position, Hessian, and mass differences are always reduced-model differences from the full Wannier source computed by the identical algorithm.

5.6 Comprehensive conduction analysis

The final identified sector-minimum coordinates from the frozen ladder are reused without reoptimization. For every model and valley, Cartesian gradients and Hessians are recomputed with $h \in \{0.001, 0.002, 0.005, 0.01\}$ Å^{-1} using Eq. (68); 0.002 Å^{-1} is primary. Define the curvature

conversion constant

$$C = \frac{\hbar^2}{m_e}\left(\frac{1\ \text{eV}}{1.602176634\times10^{-19}\ \text{J}}\right)\left(10^{20}\frac{\text{Å}^2}{\text{m}^2}\right) \simeq 7.619964\ \text{eV Å}^2. \tag{69}$$

The dimensionless inverse-mass and mass tensors are

$$M^{-1} = K/C, \qquad M = CK^{-1}. \tag{70}$$

If m_{v1}, m_{v2}, m_{v3} are a valley's principal masses, the reported one-valley summary conventions are

$$m_{\text{DOS},v} = (m_{v1}m_{v2}m_{v3})^{1/3}, \qquad m_{\text{cond},v} = \frac{3}{m_{v1}^{-1}+m_{v2}^{-1}+m_{v3}^{-1}}. \tag{71}$$

For six valleys,

$$m_{\text{DOS},6} = \left[\sum_v (m_{v1}m_{v2}m_{v3})^{1/2}\right]^{2/3}, \tag{72}$$

$$\overline{M^{-1}} = \frac{1}{6}\sum_v M_v^{-1}, \qquad m_{\text{cond},6} = \frac{3}{\operatorname{tr}\overline{M^{-1}}}. \tag{73}$$

These are declared geometric conventions derived from represented curvatures. They do not include occupations, scattering, temperature, carrier density, or a transport solution. A separate implementation-level audit, importing no silicon producer, recomputed every single-valley mass, the six-valley DOS convention, the equal-population average inverse-mass tensor, and the resulting scalar conductivity convention directly from the retained primary tensors. The maximum absolute difference was 6.22×10^{-15} in the reported dimensionless mass units. This confirms the formulas and six-valley degeneracy convention, but it does not independently reconstruct the valleys or Hessians that supplied those tensors. Gradient residuals and mass variation over all four stencils are retained to separate reduction effects from finite-difference sensitivity.

5.7 Gamma valence cluster and ordered-branch diagnostic

At Gamma, zero-based bands 1–3 define the upper valence cluster. If X_{W} and X_{r} are orthonormal source and candidate cluster frames, the singular values σ_i of $X_{\text{W}}^\dagger X_{\text{r}}$ give principal angles $\theta_i = \arccos\sigma_i$. The projector discrepancy is $\left\|X_{\text{r}}X_{\text{r}}^\dagger - X_{\text{W}}X_{\text{W}}^\dagger\right\|_F$. Cluster energies, centroid, splitting, singular values, projector distance, and principal angles are all reported. These quantities separate preservation of the three-dimensional subspace from preservation of the operator within that subspace.

As an explicit diagnostic of why scalar branch masses fail, the three ordered cluster eigenvalues are sampled along [100], [110], and [111]. Five positive points and their negative partners are used inside each half-window $Q \in \{0.005, 0.01, 0.02\}$ Å^{-1}. Each ordered branch is fitted to

$$E_n(q) = c_n + a_n q^2 + b_n q^4, \tag{74}$$

with apparent curvature $2a_n$ and apparent hole mass $-C/(2a_n)$ only when $2a_n < 0$. Missing masses correspond to positive fitted curvature; they are not clipped. Because sorting can exchange branch identity at a degeneracy, these fits are retained solely as instability diagnostics.

5.8 Transported three-band quadratic Hamiltonian

The primary valence object never assigns an individual band at Gamma. For each model, let X_0 be the Gamma cluster frame. At displaced momentum $\mathbf{q}$, diagonalization gives cluster eigenvalues $\boldsymbol{\epsilon}(\mathbf{q})$ and frame $X(\mathbf{q})$. Write

$$X_0^\dagger X(\mathbf{q}) = L(\mathbf{q})\Sigma(\mathbf{q})R(\mathbf{q})^\dagger, \qquad Q(\mathbf{q}) = L(\mathbf{q})R(\mathbf{q})^\dagger. \tag{75}$$

The represented displaced cluster Hamiltonian in Gamma coordinates is

$$\mathcal{H}(\mathbf{q}) = Q(\mathbf{q})\,\mathrm{diag}[\boldsymbol{\epsilon}(\mathbf{q})]Q(\mathbf{q})^{\dagger}. \tag{76}$$

The largest principal angle encountered during sampling is retained as a transport condition diagnostic. The even response

$$\Delta\mathcal{H}_{+}(\mathbf{q}) = \frac{\mathcal{H}(\mathbf{q}) + \mathcal{H}(-\mathbf{q})}{2} - \mathcal{H}(0) \tag{77}$$

is fitted by complex linear least squares to

$$\begin{aligned} H_{\mathrm{eff}}(\mathbf{q}) = H_{\mathrm{eff}}(0) &+ q_x^2 B_{xx} + q_y^2 B_{yy} + q_z^2 B_{zz} \\ &+ 2q_x q_y B_{xy} + 2q_x q_z B_{xz} + 2q_y q_z B_{yz}. \end{aligned} \tag{78}$$

The sample directions are the 13 unoriented rays obtained from all nonzero vectors in $\{-1,0,1\}^3$ after identifying $\mathbf{n}$ and $-\mathbf{n}$. For each half-window $Q \in \{0.005, 0.01, 0.02\}$ Å^{-1}, four radii $Q/4, Q/2, 3Q/4, Q$ are used on every ray. Thus each fit uses $N_q = 52$ even matrix observations. Each of the six fitted 3×3 matrices is explicitly Hermitian symmetrized. The RMS fit residual is

$$\left[\frac{1}{3N_q}\sum_{a=1}^{N_q} \|\Delta\mathcal{H}_{+}(\mathbf{q}_a) - \Delta\mathcal{H}_{\mathrm{fit}}(\mathbf{q}_a)\|_F^2\right]^{1/2}. \tag{79}$$

For a unit direction $\mathbf{n}$, define

$$B(\mathbf{n}) = n_x^2 B_{xx} + n_y^2 B_{yy} + n_z^2 B_{zz} + 2n_x n_y B_{xy} + 2n_x n_z B_{xz} + 2n_y n_z B_{yz}. \tag{80}$$

The curvature channels are the three eigenvalues of $2B(\mathbf{n})$. Negative channels are converted to apparent non-SOC hole masses by $m_h/m_e = -C/\kappa$. For source-relative coefficient comparison, the candidate Gamma frame is first polar-aligned to the source Gamma frame; the same unitary conjugates all six candidate B_{ij}. The reported joint coefficient error is

$$\eta_B = \frac{[\sum_{ij} \left\| B_{ij}^{\mathrm{r}} - B_{ij}^{\mathrm{W}} \right\|_F^2]^{1/2}}{[\sum_{ij} \left\| B_{ij}^{\mathrm{W}} \right\|_F^2]^{1/2}}. \tag{81}$$

Window sensitivity compares each coefficient field with the primary $Q = 0.01$ Å^{-1} field by the same normalized joint norm.

Finally, direct gaps are evaluated at Gamma, six conventional X points $\pm(2\pi/a)\mathbf{e}_i$, and eight conventional L points $(\pi/a)(\pm1, \pm1, \pm1)$ after conversion to reduced primitive coordinates. The minimum, maximum, and spread over equivalent X or L points are retained. A separate Gamma-origin 32^3 mesh supplies every ordered-band minimum, maximum, bandwidth, VBM, CBM, and indirect gap for all four frozen models.

6 Silicon results

Evidence and reference boundary. Every silicon result in this section is an authenticated, producer-checked development calculation pending independent reconstruction; the section is not an independently verified material demonstration. Reduction fidelity is measured against the full rank-10 Wannier source, source adequacy is assessed separately against the revealed DFT parent, and no comparison with experiment is made.

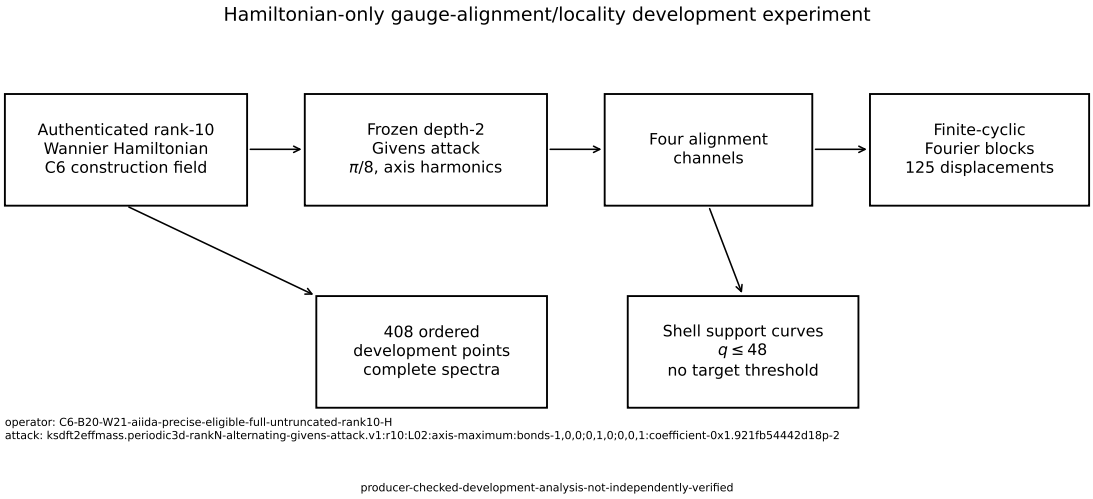


Figure 9: Schematic of the rank-10 silicon gauge and finite-support experiment. The source, internal attack, alignment channels, finite Fourier route, and support decisions are separate stages with separate error ownership.

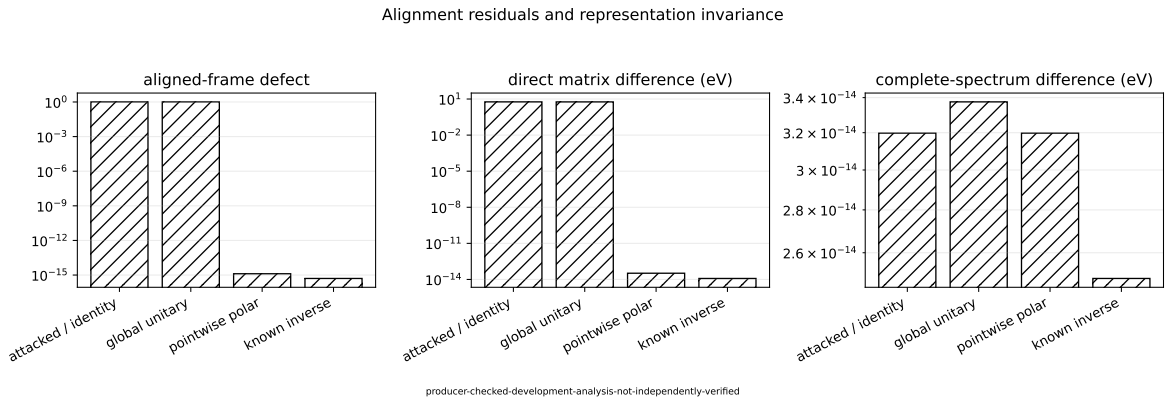


Figure 10: Alignment and complete-spectrum invariance for the rank-10 silicon attack. The nonzero identity/global residual is caused by the restricted alignment family; the complete fields remain related by the generating momentum-dependent unitary.

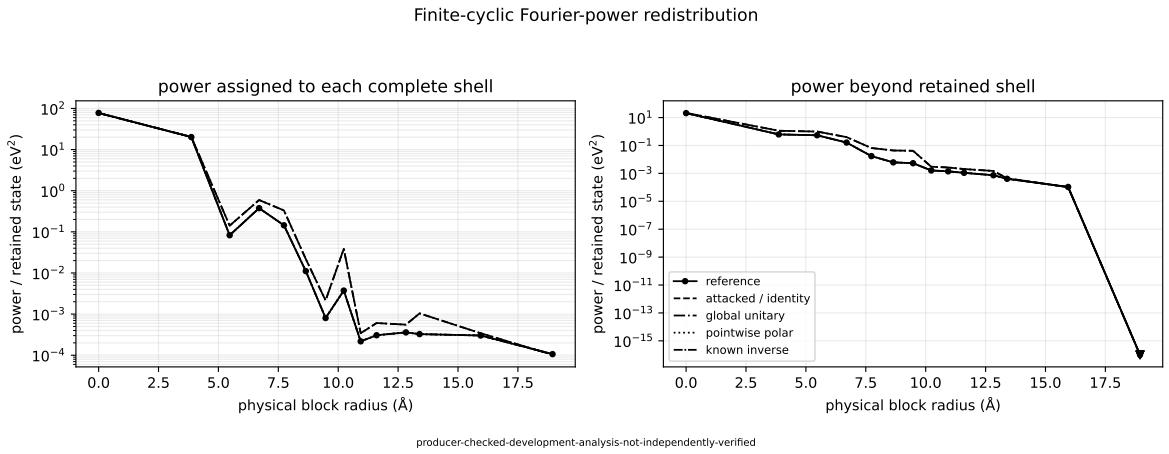


Figure 11: Rank-10 Fourier-power redistribution under a spectrum-preserving internal attack. Block power is plotted against physical radius; the producer figure distinguishes the conversion channel, but its exact removed-power totals were not serialized in the retained machine-readable result.

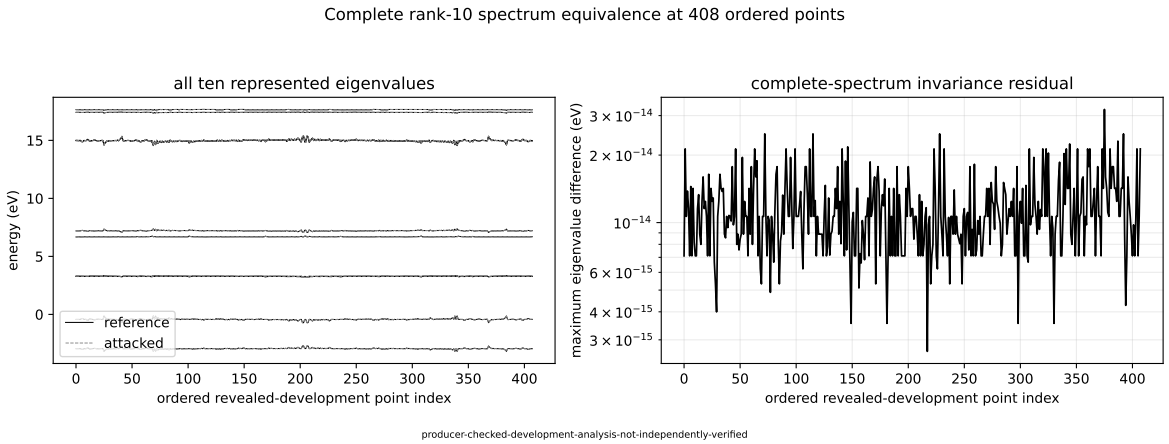


Figure 12: Complete reference and attacked spectra on the ordered 408-point inventory. The horizontal coordinate is an ordered inventory, not an interpolated band path.

6.1 Complete spectrum is invariant while locality changes

Within that producer-checked scope, the maximum complete-spectrum difference under the frozen attack is 3.1974×10^{-14} eV. Identity and one-global-unitary alignment leave maximum frame defects near 1.021 and 1.020. Pointwise polar and known-inverse controls reduce the defect to approximately 1.29×10^{-15} and 5.06×10^{-16}.

The reference and identity-attacked non-onsite finite-block powers are 20.7574 and 21.5057 eV^2 per retained state. The finite-cyclic interpolation maximum sorted-band errors are approximately 0.3470 and 0.3867 eV. These interpolation floors are not attack-induced complete-spectrum errors; they belong to the finite construction and conversion route. The producer used the 91 removed Nyquist-boundary coefficients as a separate conversion channel when rendering the original diagnostic, but the retained JSON serialized only the 125 ordinary-mode powers, not the 216 cyclic matrices or the two removed-power totals. Exact numerical conversion-removed powers are therefore unavailable from the authenticated retained result and are not reconstructed or quoted here. This archival omission remains part of the independent Fourier-reconstruction blocker.

The observed non-dominated support-error sets change between representations. This is the higher-rank material-derived counterpart of the rank-two range-inflation mechanism: complete

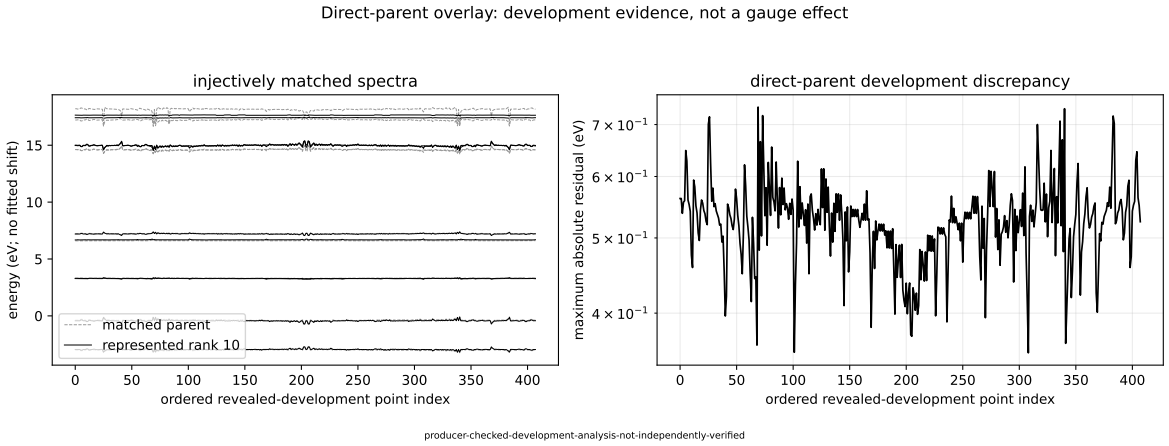


Figure 13: Direct-parent overlay for the represented source and gauge-equivalent comparison. The displayed parent discrepancy belongs to source adequacy and remains distinct from the exactly spectrum-preserving internal attack.

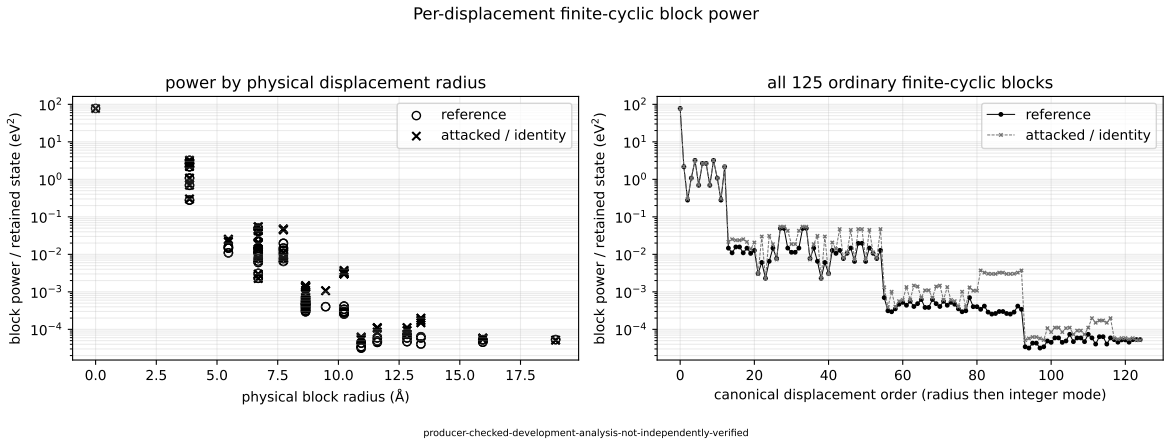


Figure 14: Displacement-resolved block power in the converted finite representation. Individual block norms reveal redistribution that can be hidden by a single total non-onsite-power scalar.

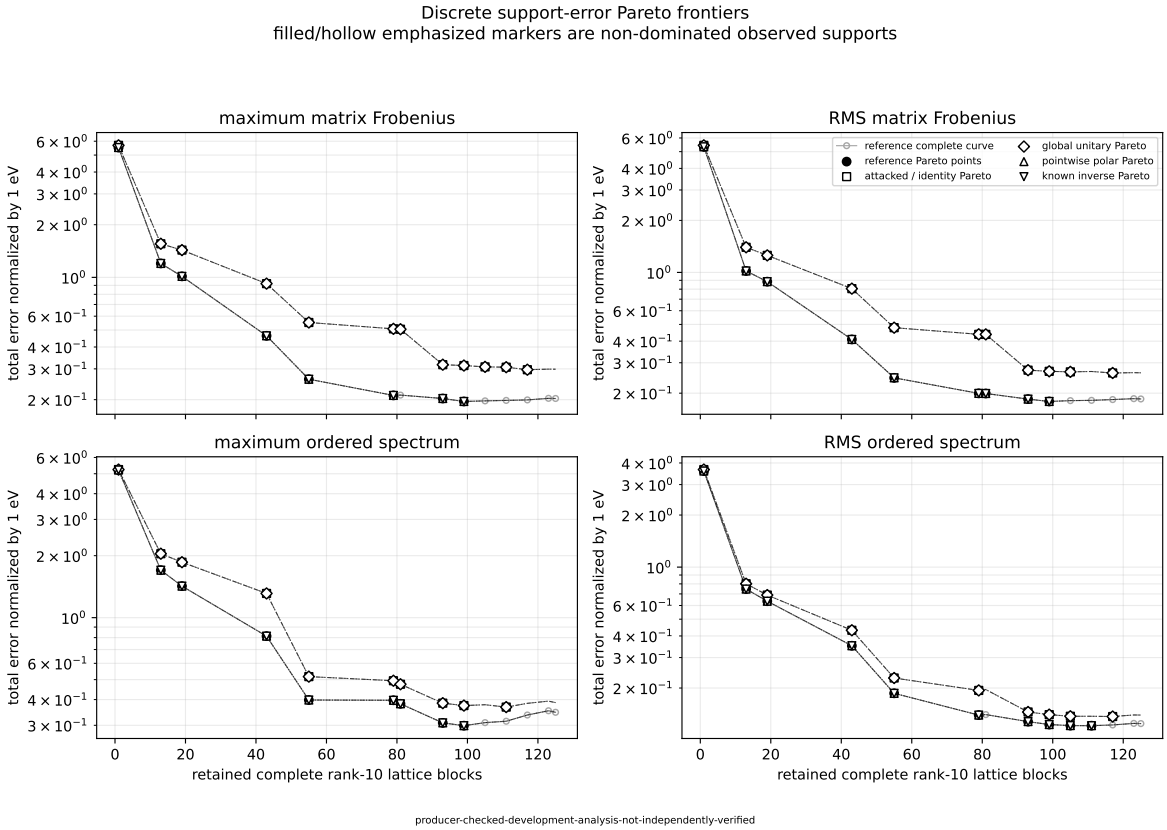


Figure 15: Observed discrete rank-10 non-dominated support-error sets. Markers identify the non-dominated complete supports found on the exhaustive declared support inventory. Lines guide the eye between discrete shell choices and do not define fractional supports.

spectral equivalence does not fix the finite support needed under a restricted coordinate identification. These are observed support-error frontiers over the exhaustive declared finite support inventory, not universal support bounds. Likewise, the equal-attained-error comparison is discrete and does not interpolate continuously in physical radius or support count.

6.2 Literal shell truncation has no global two-loss conflict

Among the 17 proper literal reductions, H-shell-16 minimizes both primary RMS losses:

$$(\tau_1, \tau_2) = (4.4636 \times 10^{-4}, 1.5173 \times 10^{-3}), \tag{82}$$

while retaining 371 representatives, or 96.9% of the 383-representative source. It is therefore a high-fidelity near-full-support reduction, not a compact short-range model. The tightest literal family has no global spectral–operator conflict: the same proper support minimizes both losses. This does not show that the losses are generally interchangeable. The family is nested, and both coordinates are driven by the same omitted matrix field.

The post-hoc pair $(0.1, 0.15)$ first admits 135 representatives, $(0.01, 0.02)$ first admits 313, and $(0.001, 0.002)$ first admits 371. The pair $(0.0004, 0.001)$ has no proper literal witness.

6.3 Refitting creates genuine spectral–operator arcs

The bounded refitted search retains 444 converged constructive witnesses. Fixed supports now trace distinct spectral–operator arcs. Observed minimum support changes include

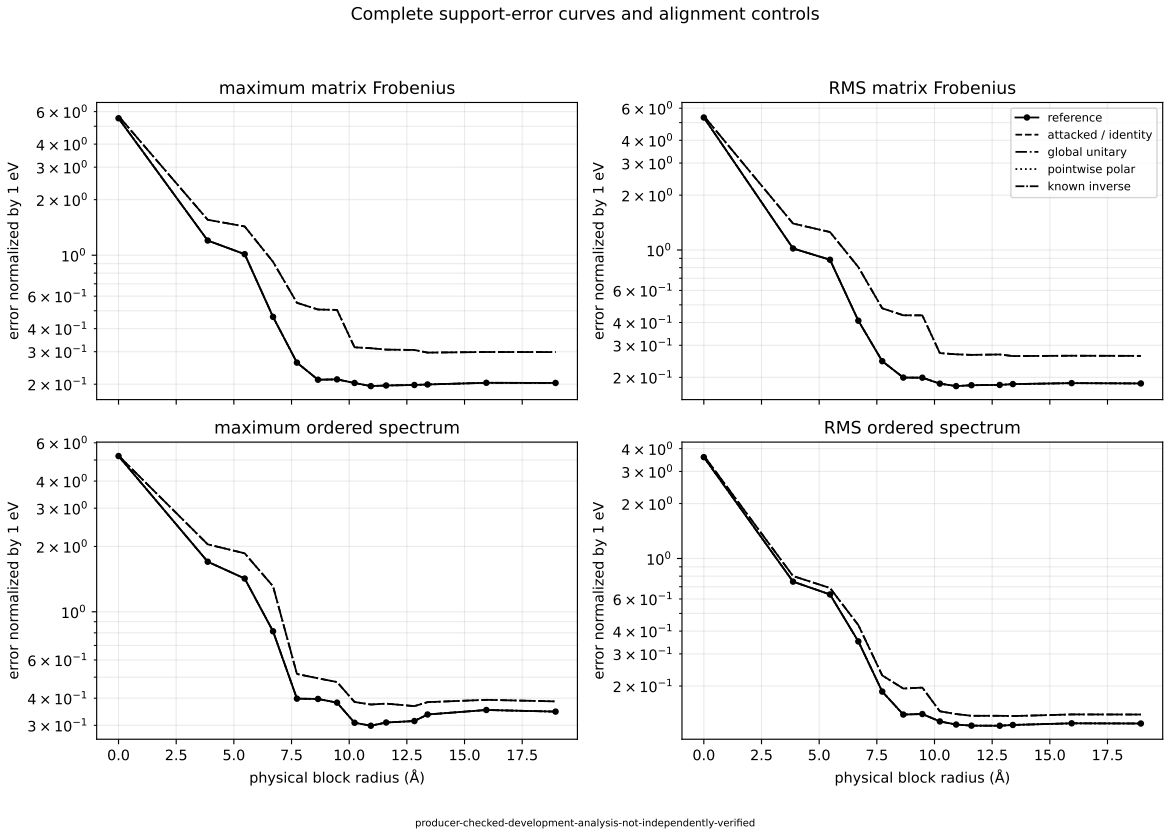


Figure 16: Complete support-error curves before non-dominated-set extraction. The curves retain dominated supports and therefore expose nonmonotonic details that a non-dominated-only summary removes.

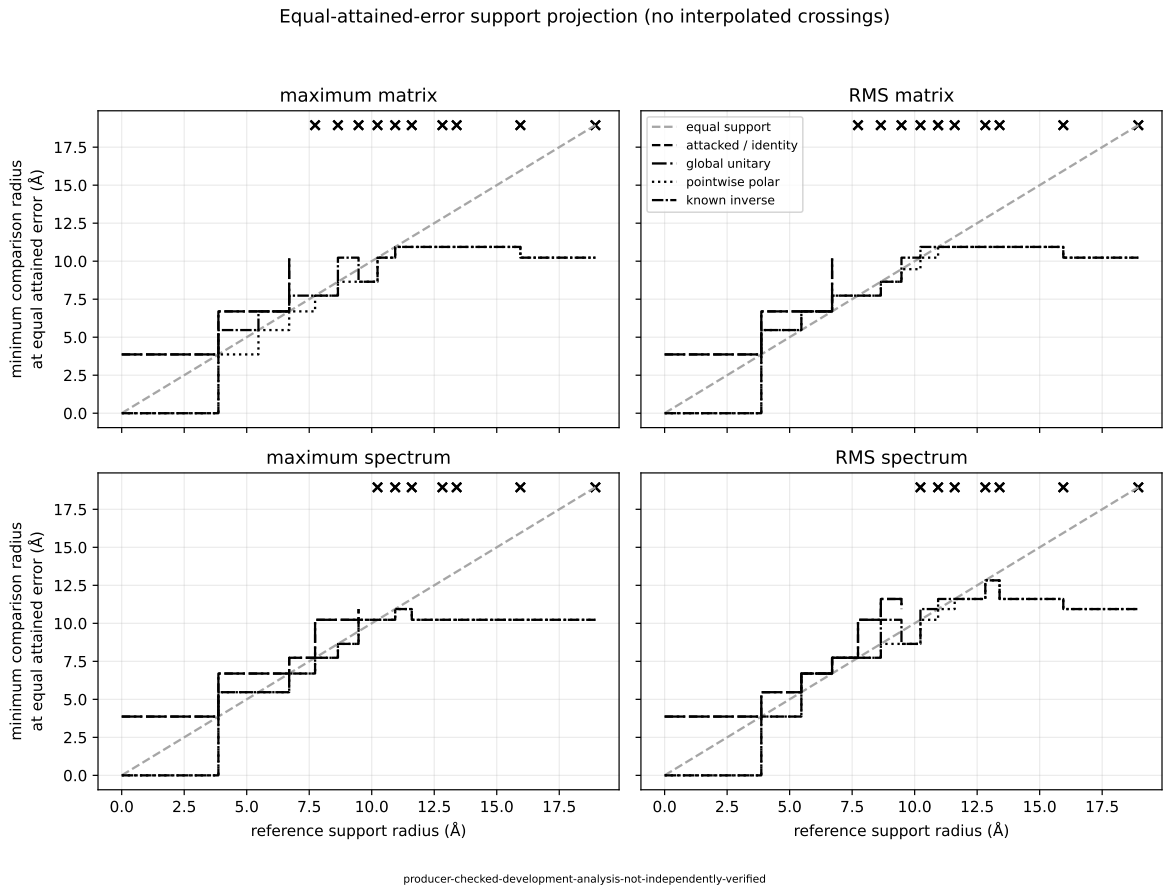


Figure 17: Equal-attained-error support comparison for the two rank-10 representations. The comparison uses attained discrete errors and avoids interpreting line segments as continuously adjustable support.

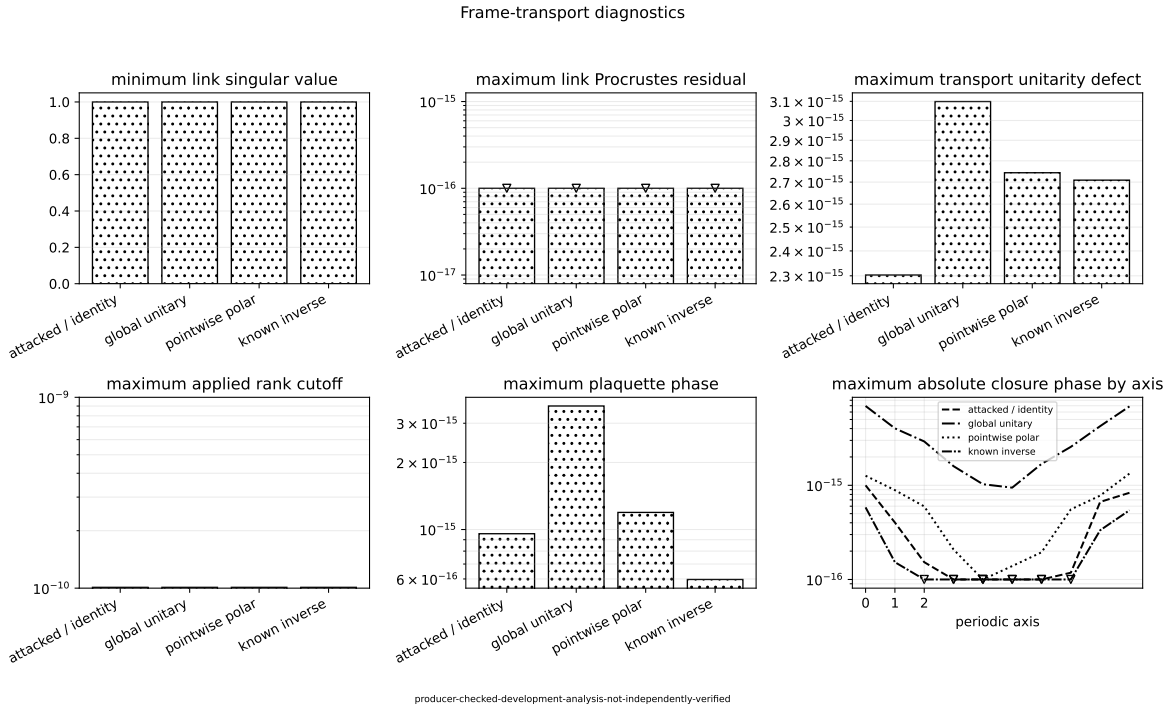


Figure 18: Transport and sewing diagnostics for the rank-10 alignment channels. Closure and plaquette phases are numerical sewing diagnostics and are not assigned a topological interpretation.

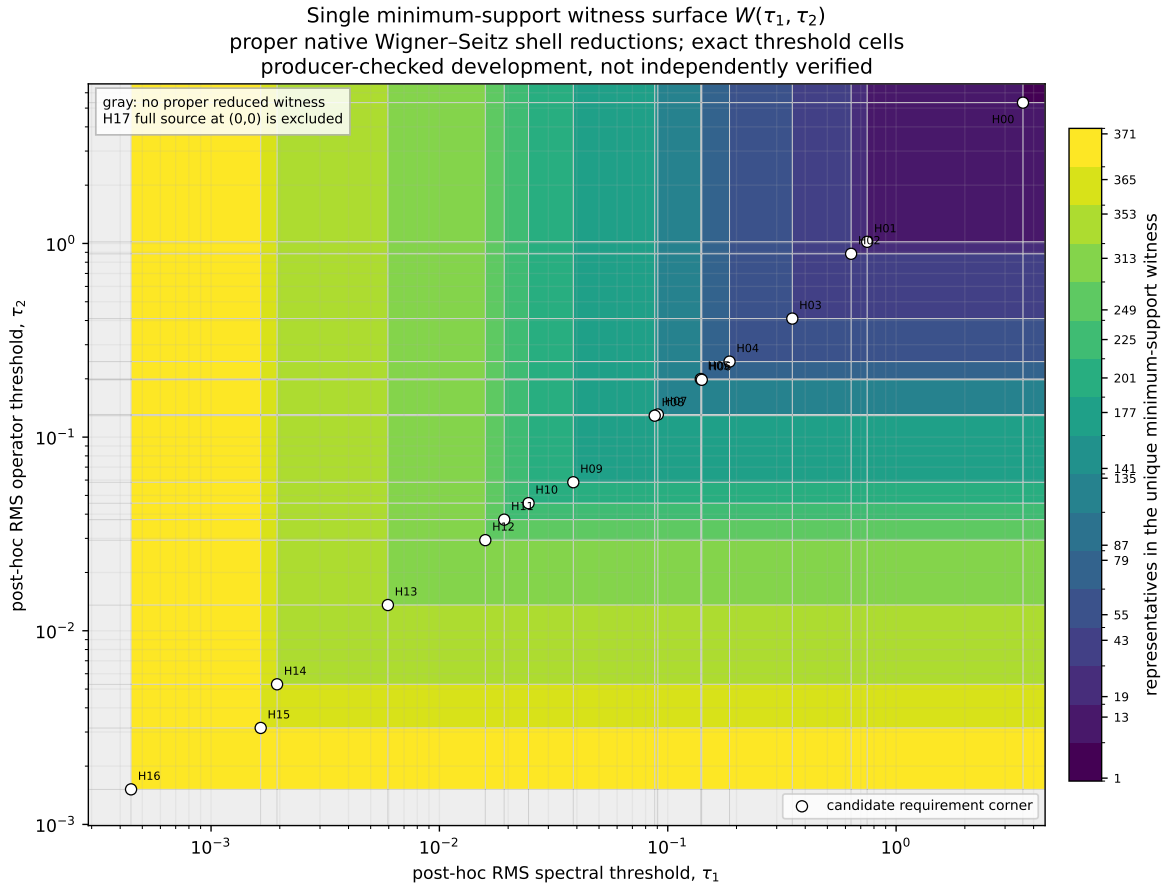


Figure 19: Minimum-support surface for literal native-shell deletion. The full 383-representative source is excluded from reduced-family feasibility. Threshold coordinates are descriptive post-hoc readings, not acceptance limits.

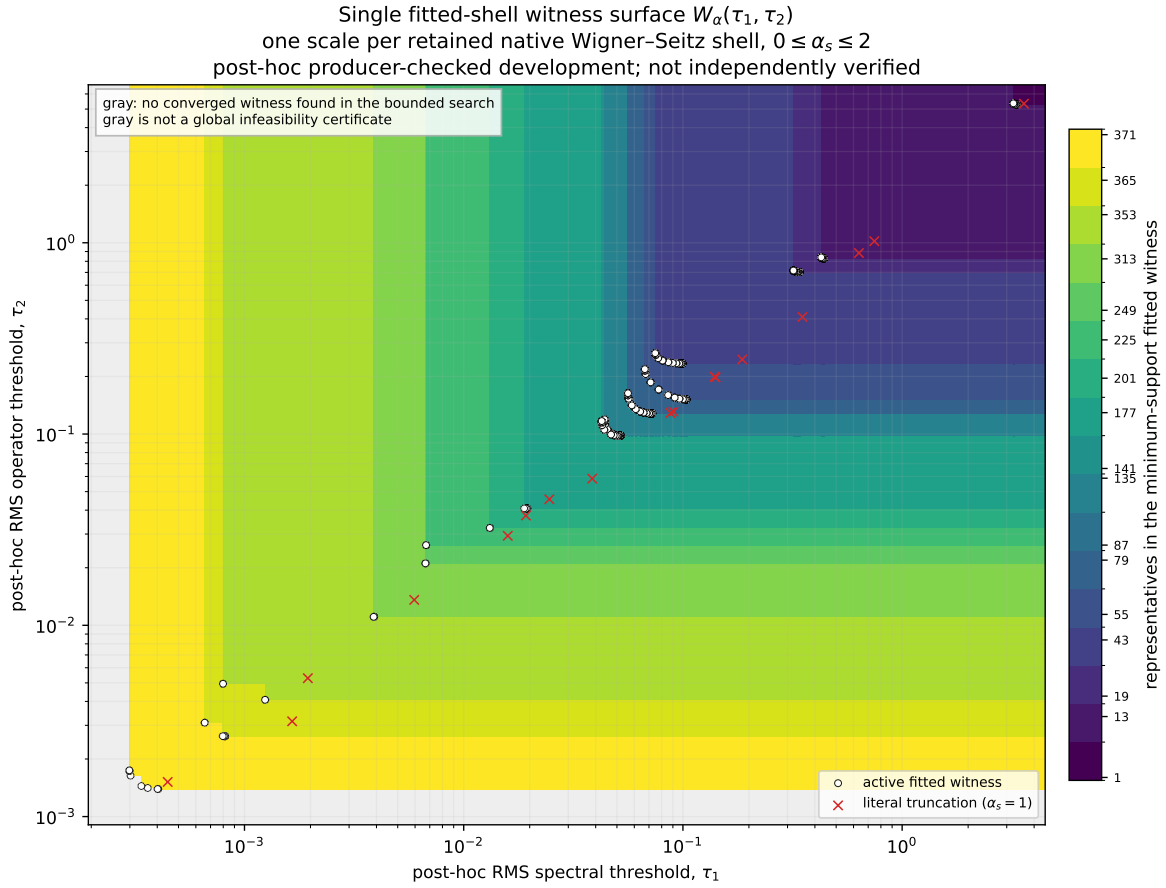


Figure 20: Minimum-support surface over converged refitted shell witnesses. White circles mark active fitted witnesses and red crosses mark literal shell models. Gray cells mean no converged witness was found by the bounded search.

(τ_1, τ_2)	Literal representatives	Refitted representatives
$(1.0, 1.0)$	19	13
$(0.1, 0.15)$	135	79
$(0.05, 0.1)$	177	135
$(0.02, 0.04)$	225	201
$(0.01, 0.02)$	313	313
$(0.001, 0.002)$	371	371

Within H-shell-16, an operator-oriented witness reaches approximately $(4.04 \times 10^{-4}, 1.394 \times 10^{-3})$, while a spectral-oriented witness reaches $(2.98 \times 10^{-4}, 1.745 \times 10^{-3})$. The same support therefore exhibits a real tradeoff after refitting. Neither reported H-shell-16 witness has an active coefficient bound; all retained bound-active records occur only on H-shell-02, H-shell-06, or H-shell-08. The optimization remains constructive; gray cells mean that the bounded multistart search found no witness and do not certify infeasibility or global optimality.

6.4 Fitted-point errors understate broader behavior

The frozen fidelity ladder gives the following comparison: The ratio columns are computed from the full binary64 records before the displayed RMS values are rounded; direct division of the printed values can therefore differ in the last shown digit.

Sampled operator RMS agrees with exact Parseval values to at most 5.56×10^{-17} eV. The 12^3 to 32^3 spectral changes remain below about 1.4%, so the dense values are stable over the

Table 3: Fitted-inventory, dense-mesh, and exact Parseval RMS errors for the frozen silicon fidelity ladder.

	Spectral RMS (eV)			Operator RMS (eV)		
Model	408	32^3	Ratio	408	Parseval	Ratio
H-shell-05	0.07242	0.18196	2.51	0.12814	0.26004	2.03
H-shell-10	0.01320	0.02785	2.11	0.03229	0.05956	1.84
H-shell-16	0.000404	0.000657	1.62	0.001394	0.001747	1.25

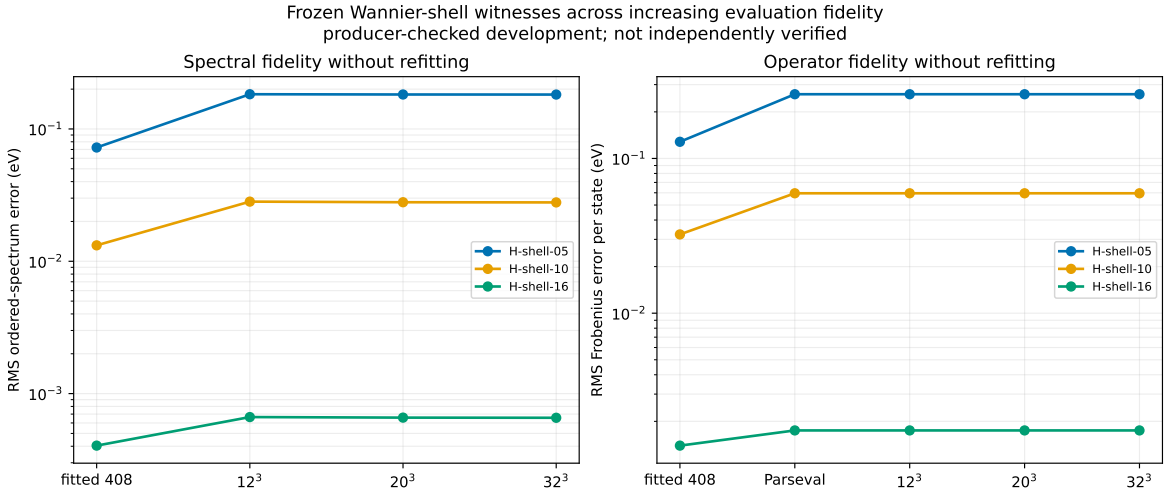


Figure 21: Frozen-model fidelity ladder from fitted points to exact and dense evaluations. Supports and shell scales are unchanged at every level.

tested sequence. The discrepancy is a real distinction between the fitted inventory and broader evaluation. In particular, H-shell-10 is not a reliable band-edge reduction: its global dense spectral RMS is only 0.02785 eV, yet a principal conduction mass changes by 53.2%.

6.5 Conduction valleys and masses are stricter tests

Relative to the full Wannier source, the identified conduction-sector minima for H-shell-05 change the indirect gap by -65.77 meV, move by as much as 0.09481 Å^{-1}, change a Hessian by 29.9%, and change a principal mass by 67.7%. H-shell-10 changes the gap by -34.09 meV and a principal mass by 53.2%, despite a 32^3 all-band spectral RMS below 0.028 eV. H-shell-16 changes the gap by -0.503 meV, the maximum identified-minimum position by 0.002758 Å^{-1}, the maximum Hessian by 0.259%, and the maximum principal mass by 2.54%. These are the lowest successful matching minima from six signed-axis sectors, using eight 32^3 coarse starts plus the source coordinate, bounded refinement in a $[-0.12, 0.12]^3$ Å^{-1} Cartesian box, 240 iterations, function tolerance 10^{-14}, and gradient tolerance 10^{-10}. They are minima found by the declared search, not certified global Brillouin-zone minima.

The six-valley DOS-mass convention of the source is $1.4764 m_e$, and its equal-population conductivity-mass convention is $0.25063 m_e$. H-shell-05 changes them by 16.9% and 20.4%; H-shell-10 by 22.9% and 6.90%; H-shell-16 by 0.333% and 0.0216%. The DOS summary is nonmonotone from H-shell-05 to H-shell-10 because it is a nonlinear geometric combination: opposing principal-mass errors in H-shell-05 partially cancel, whereas the H-shell-10 shifts are more coherently signed. This cancellation does not imply better tensor fidelity for H-shell-05. These quantities are curvature conventions only, not carrier-transport predictions. Across all models, the maximum principal-mass spread over the four finite-difference steps is below 0.172%, so stencil variation does not explain the moderate-support errors.

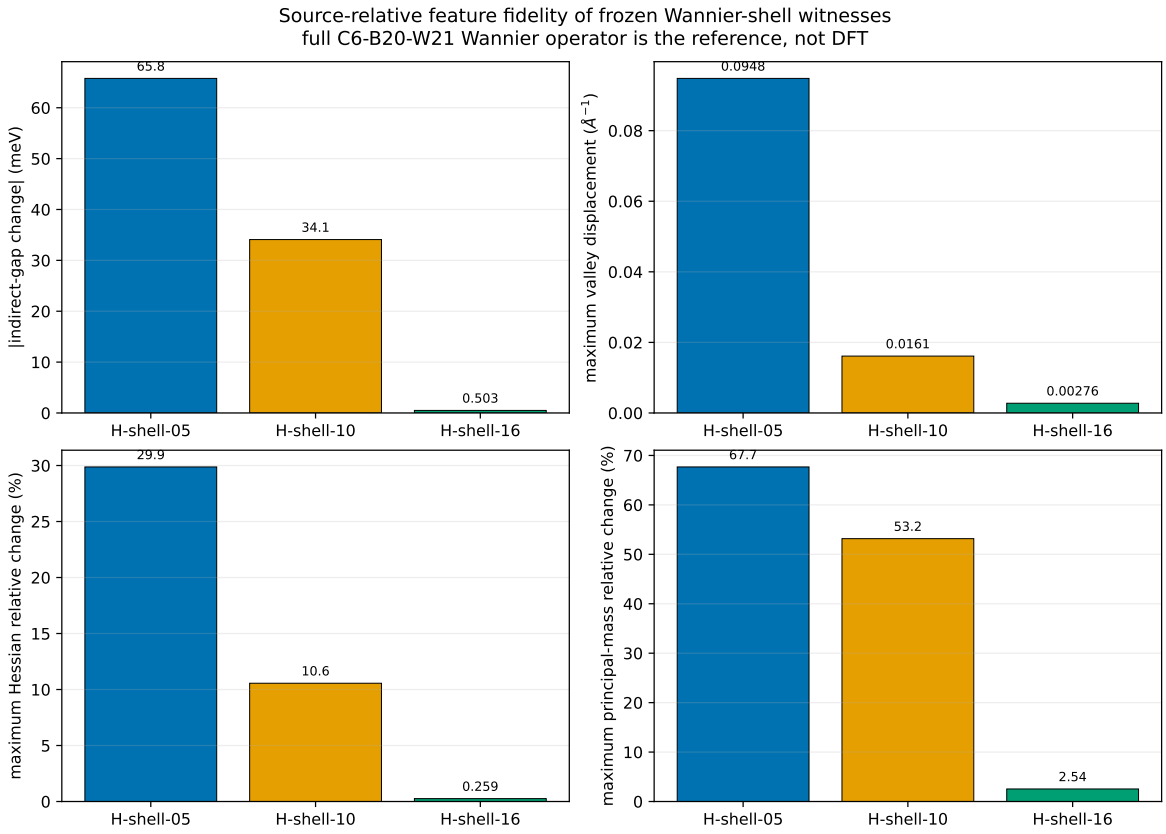


Figure 22: Source-relative conduction-edge fidelity of the frozen models. The reference is the full Wannier source, not the revealed DFT parent.

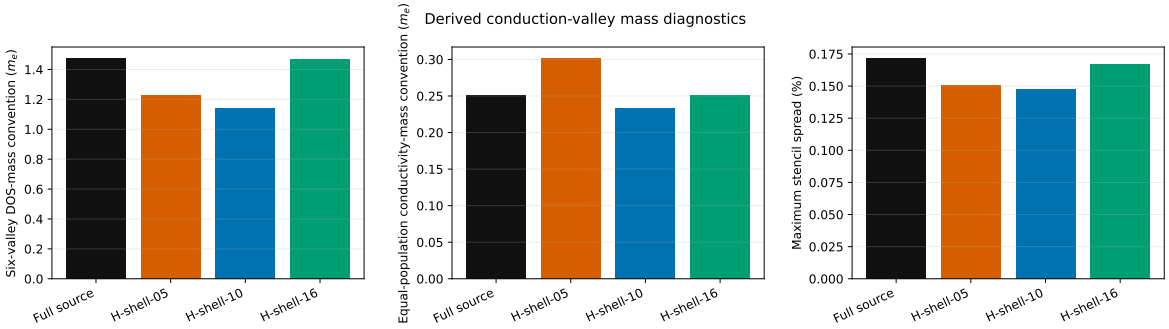


Figure 23: Derived conduction-valley mass summaries and finite-difference sensitivity. The DOS and conductivity masses are declared curvature conventions and are not carrier-density or transport calculations.

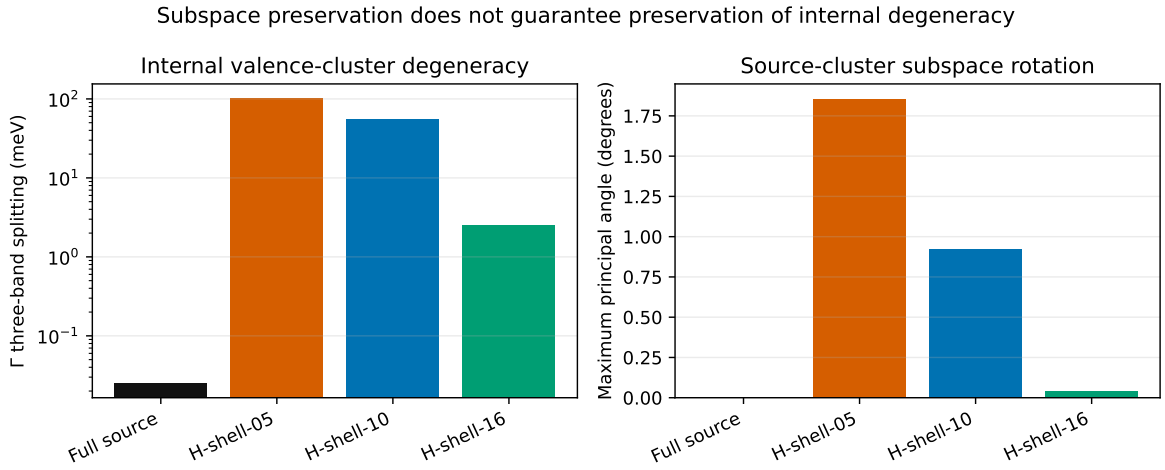


Figure 24: Gamma valence-cluster splitting and source-subspace rotation. Internal degeneracy and subspace preservation are independent fidelity conditions.

6.6 The valence manifold requires a multiband operator

The source Gamma three-band splitting is only 0.02495 meV. H-shell-05 increases it to 102.175 meV, H-shell-10 to 54.806 meV, and H-shell-16 to 2.496 meV. The corresponding maximum principal angles from the source cluster are only 1.856, 0.925, and 0.0420 degrees. A candidate can therefore preserve a recognizable three-dimensional subspace while badly damaging the operator inside that subspace. In short, subspace fidelity does not imply internal Hamiltonian fidelity.

Independently ordered source branches do not produce stable hole masses. The upper ordered branch is not hole-like for every tested window in any of the three directions, and finite ordered-branch masses can change by about 75.5%. H-shell-05 and H-shell-10 appear more stable only because they artificially split the manifold.

The transported quadratic operator avoids that branch assignment. At the primary fit half-window $Q = 0.01$ Å^{-1}, the source apparent channel masses are

Direction	Channel 1	Channel 2	Channel 3
[100]	0.1411	0.2771	0.3015
[110]	0.09690	0.3119	2.923
[111]	0.08770	0.6971	0.7722

in electron-mass units. These are non-SOC curvature-channel eigenvalues, not physical HH, LH, or split-off masses.

After cluster alignment, H-shell-05, H-shell-10, and H-shell-16 have quadratic-coefficient errors of 9.54%, 5.88%, and 0.279%, with maximum channel-mass changes of 38.9%, 19.7%, and 4.53%. Fit-window changes in the joint coefficient field remain below about 0.41%, and primary fit residuals are approximately 5–6 $\times 10^{-7}$ eV RMS.

6.7 High-symmetry equivalence and bandwidths

The full source's direct-gap spread among nominally equivalent X points is 0.00366 meV and among L points is 0.0239 meV. H-shell-05 increases these spreads to 150.1 and 68.7 meV; H-shell-10 to 36.7 and 63.9 meV; H-shell-16 to 0.427 and 2.97 meV. Moderate shell refitting can therefore introduce numerical inequivalence among points that are symmetry-equivalent in the declared source construction, through cancellations among retained blocks. These spreads do not independently prove exact physical silicon symmetry or an experimentally established symmetry violation.

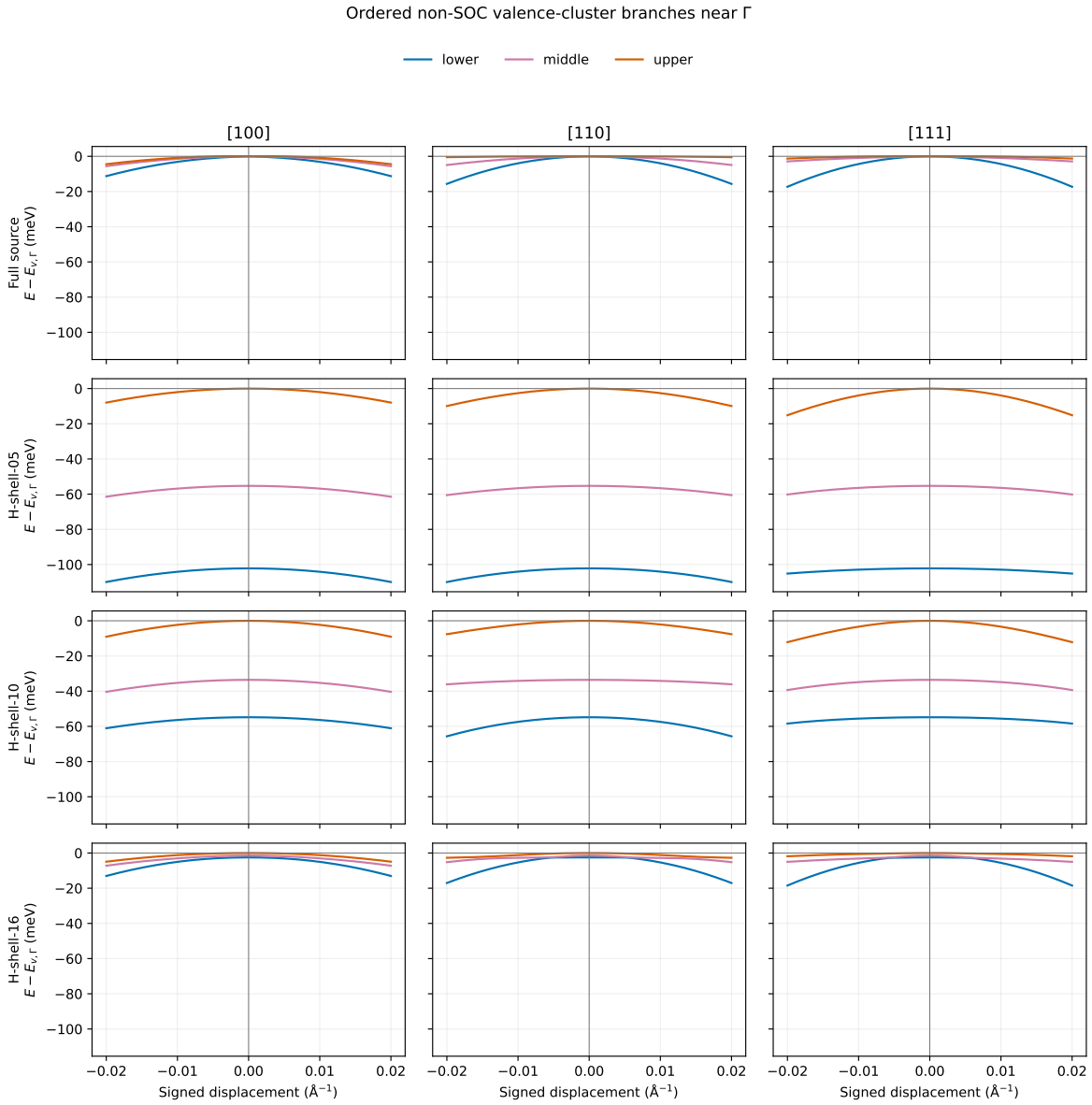


Figure 25: Directional ordered valence-cluster dispersions near Gamma. The curves are diagnostics of branch ordering and degeneracy lifting, not a physical assignment of heavy-hole, light-hole, and split-off bands.

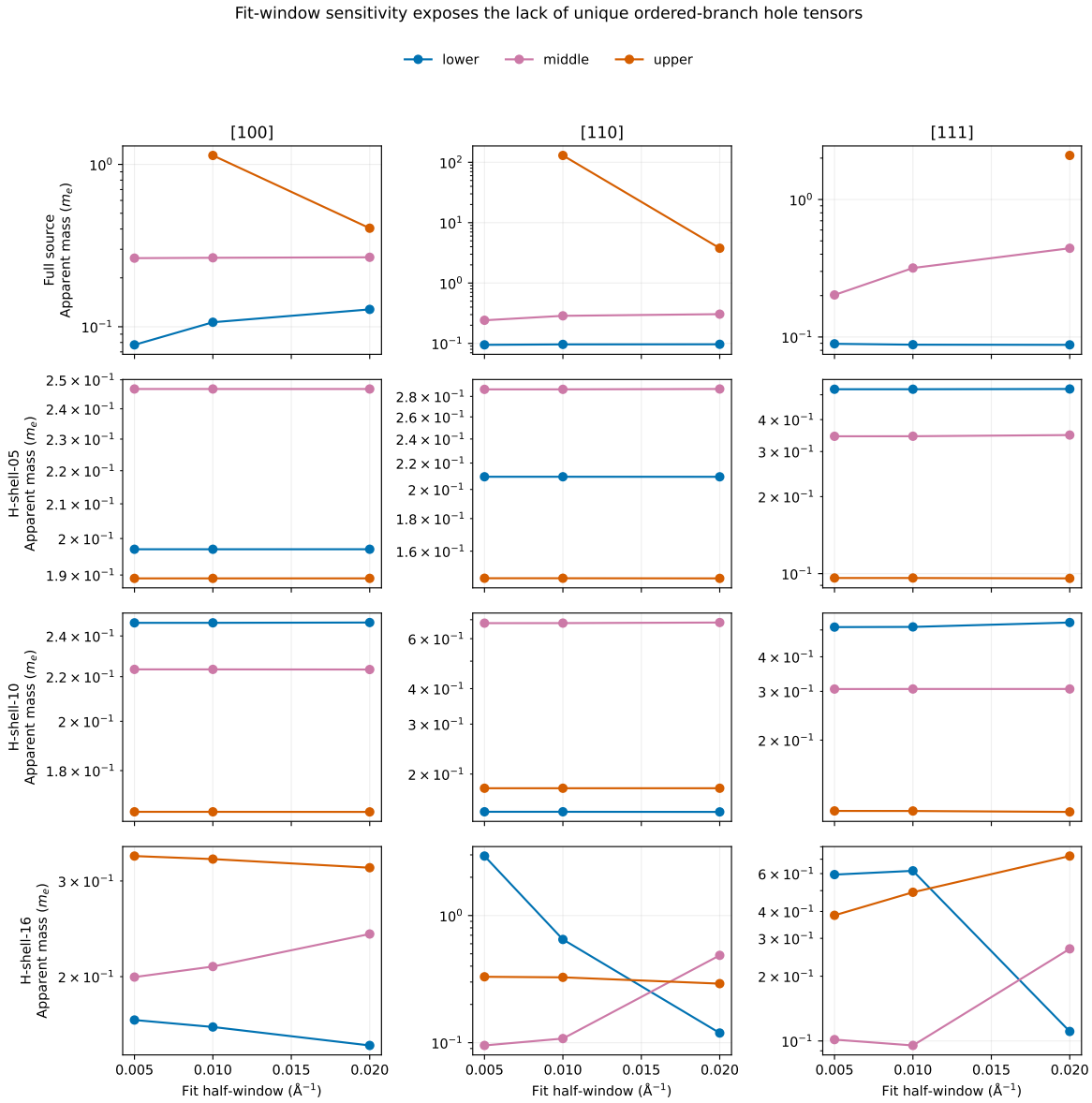


Figure 26: Instability of independently ordered valence-branch mass fits. Window dependence and loss of hole-like curvature motivate replacing scalar branch masses with a transported multiband quadratic operator. Missing curves denote positive fitted curvature, for which no apparent hole mass is reported.

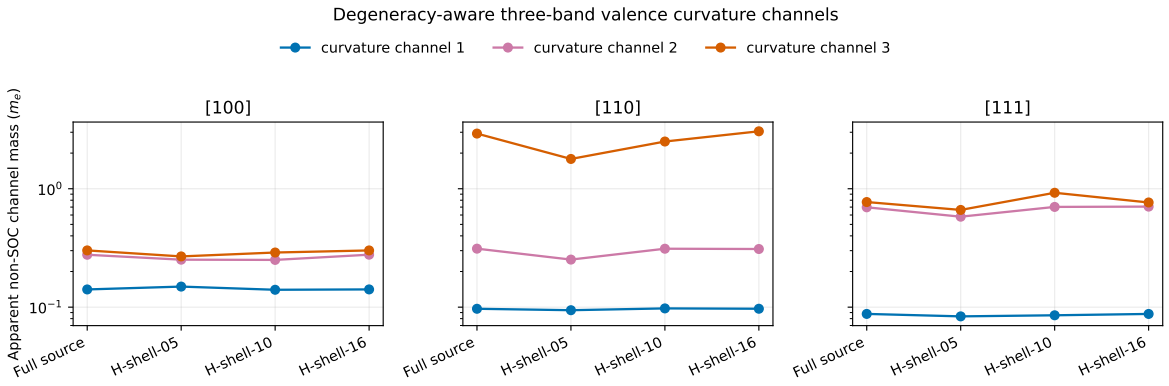


Figure 27: Degeneracy-aware non-SOC valence curvature channels. The channels are eigenvalues of the direction-contracted quadratic matrix and are not assigned SOC hole labels.

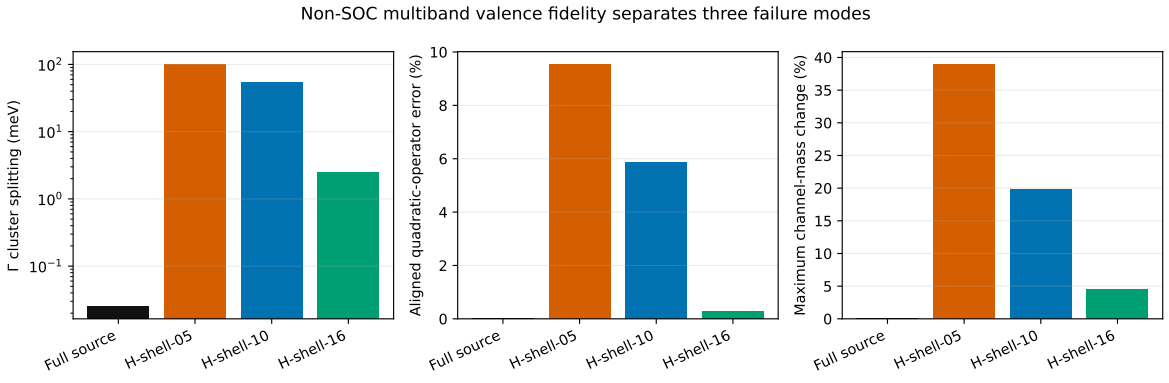


Figure 28: Summary of valence degeneracy, quadratic-operator, and curvature consequences. The three panels separate failure modes that a single band RMS cannot represent.

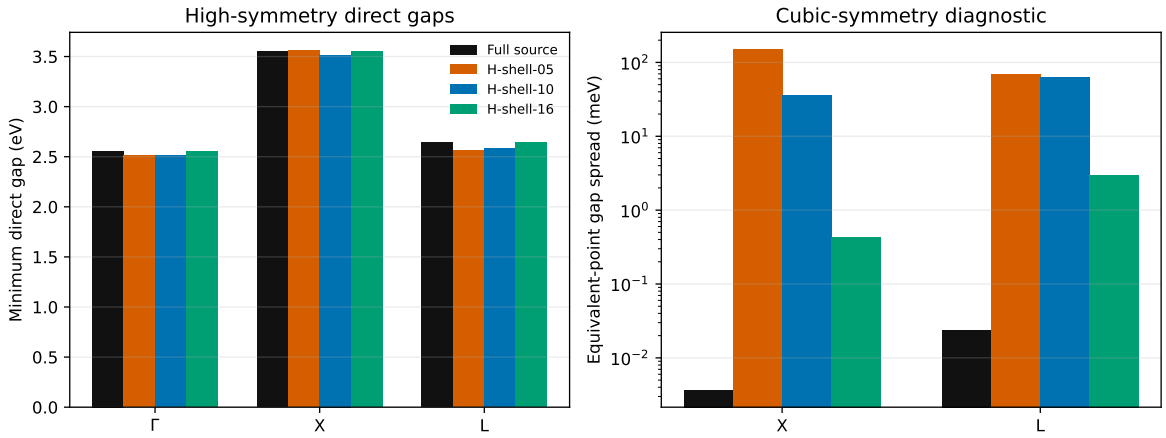


Figure 29: High-symmetry direct-gap and equivalent-point diagnostics. The spread characterizes the represented models; it is not an independent validation of exact crystal symmetry.

On the independent 32^3 mesh, all four valence maxima remain at Gamma. Maximum ordered-band bandwidth changes are approximately 354 meV, 81.1 meV, and 3.14 meV for H-shell-05, H-shell-10, and H-shell-16.

7 Discussion

7.1 Related mechanism under distinct evidential roles

The rank-two theorem and rank-10 silicon attack differ greatly in dimension, origin, geometry, and numerical complexity. The former proves an identity in a restricted family; the latter is a source-specific computational stress test. They nevertheless isolate the same logical fact. Pointwise unitary equivalence fixes complete spectra and leaves the represented abstract operator unchanged under the generating map. It does not fix the matrix field under a restricted coordinate identification, and it does not fix the Fourier block distribution.

In the flat rank-two case, the mismatch has an exact closed form and equals induced non-onsite power. In the dispersive and rank-10 cases, no universal equality is claimed. Instead, block power, support curves, transport diagnostics, interpolation, and truncation consequences are evaluated independently. This is a stronger scientific position than extending the exact theorem by analogy: the theorem identifies the mechanism, and the calculations test which parts survive when its assumptions are relaxed.

7.2 Representation efficiency is not physical inequivalence

The reference and attacked complete fields are physically equivalent within the fixed retained space under the known momentum-dependent unitary. The reported residuals arise because the comparison admits a smaller family: one global real rotation in the rank-two theory, or identity/global/polar/oracle channels in the rank-10 application. Likewise, range inflation appears only after a support family and accuracy criterion are chosen.

The phrase "less local" therefore means more expensive in the declared representation under the declared finite-support approximation. It does not mean that the complete gauge-related Hamiltonian describes different physics. This distinction is essential when locality is used as a computational resource or model-selection coordinate.

7.3 Literal deletion and refitting answer different questions

Literal nested shell deletion asks what happens when the source coefficients are kept unchanged inside a radius and set exactly to zero outside it. Its near-linear spectral–operator requirement curve is a property of a one-coordinate nested family. The result that H-shell-16 minimizes both losses does not prove general compatibility of spectral and operator objectives.

Refitted shells introduce additional coordinates while preserving complete native shells and the common Wannier basis. Genuine same-support tradeoff arcs then appear, and moderate post-hoc support requirements improve. The distinction is scientifically useful: the literal study measures deletion error, while the fitted study measures what can be compensated by bounded renormalization of retained shells.

At tight errors, both approaches still require near-full support within this whole-block Bravais-shell ansatz. Refitting does not turn the source into a compact 10-orbital short-range model. It supplies explicit witnesses and a more expressive finite family, not a global optimality theorem. Entrywise, orbital-pair-distance, orbital-channel, and symmetry-constrained reductions remain untested comparison families; the observed 371-of-383 requirement must not be extrapolated to them.

7.4 A fitted inventory is not a Brillouin-zone guarantee

The 408-point inventory supports exact finite statements about the loss on those points. It does not by itself support all-$\mathbf{k}$ interpolation claims. The fidelity ladder freezes the models and then asks broader questions. Parseval provides exact integrated operator RMS for the finite Fourier series. Dense meshes test spectra between the fitted points. Both reveal larger errors, particularly at moderate support.

This distinction resolves a common confusion about held-out data. Held-out points are not required to report a finite training-inventory result honestly. They are required when the claim changes from "this model fits these points" to "this model interpolates the represented Hamiltonian over a broader domain."

7.5 Band edges are structurally more demanding than global spectra

Conduction masses and valence curvature depend on derivatives. A global all-band RMS can remain moderate while a local curvature changes by tens of percent. Eigenvalue errors measure band values, effective-mass errors measure local second derivatives, and valence-cluster errors measure a matrix-valued operator and its internal structure. Neither the derivatives nor that internal nearly degenerate structure is controlled by the reported global spectral RMS without additional regularity and error-control assumptions. H-shell-10 is the clearest example: its dense spectral RMS is below 0.028 eV, yet the maximum principal electron-mass change is 53.2% and the maximum non-SOC valence curvature channel-mass change is 19.7%.

The valence analysis adds a qualitative warning. Independent ordered-band Hessians can become stable only after an incorrect reduction lifts the degeneracy. Numerical stability is not fidelity if the model has changed the state-space structure that made the original scalar quantity ill defined. A matrix-valued multiband expansion is the appropriate retained object.

7.6 What can and cannot be said about holes

The current source is scalar-relativistic and non-SOC. It supports a three-band orbital valence operator and directional curvature channels. It does not support standard physical HH/LH/split-off labels, a split-off energy, or final boron-acceptor modeling. Those claims require a spinor state space and a separately qualified SOC DFT/Wannier branch. SOC is not required for the internal conclusion that reduction damages the non-SOC valence manifold.

7.7 Why the source inadequacy matters

The strongest positive statement about H-shell-16 is that it closely preserves the full Wannier source across the tested internal channels. The strongest negative statement about H-shell-05 and H-shell-10 is that they do not. Neither statement validates the source. The revealed DFT-parent comparison already shows that the source misses the declared gap and electron-mass targets.

The one-dimensional calibration hierarchy makes the limitation operational. Exact spectral compression of a finite matrix does not validate that matrix against its continuum parent, and grid refinement at fixed box width converges to the finite-box problem rather than repairing the boundary truncation. Likewise, enlarging the hopping support at fixed Wannier rank converges to the fixed Wannier source. It cannot repair a band-window, construction-mesh, parent-model, or source/DFT discrepancy upstream. Nevertheless, logical separation of error channels does not imply causal independence. The present rank-10 source freezes states through 8 eV in a d-free sp^3 plus radial s trial space on a 6^3 mesh. Over-constrained disentanglement or limited projectability can simultaneously degrade parent band-edge fidelity and reduce gauge smoothness, thereby increasing long-range Fourier tails [30, 13]. No alternative window, $3d$-augmented projection set, or projectability-disentangled source was constructed here, so this plausible shared mechanism is a future comparison rather than an established diagnosis of the retained operator.

Keeping this separation explicit prevents a common error chain:

$$\text{small reduced/source error} \not\Rightarrow \text{small source/DFT error} \not\Rightarrow \text{small DFT/experiment error.}$$

Each arrow requires its own calculation and authority.

7.8 Slater–Koster is not the active conclusion

An earlier planned Slater–Koster study proposed compatibility inside an orthogonal ten-orbital sp^3s^* hierarchy. The attempted common-space gate found rank 9 at Gamma and one additional construction point. The joint operator fit therefore stopped before optimization. A spectral-only Slater–Koster fit remains mathematically possible, but it would answer a narrower eigenvalue question and is not the active reduction path.

The completed program instead studies compression directly in the common rank-10 Wannier basis. This avoids inventing an orbital alignment that the evidence does not support and permits finite operator, gauge, Fourier, truncation, and band-edge comparisons. The present manuscript does not claim that Slater–Koster models are impossible; it reports that the planned joint operator route was blocked and that a Wannier-only route was completed instead.

7.9 Evidence maturity

The analytic rank-two statements are mathematical results under explicit assumptions and carry the paper's central mathematical claim. The synthetic non-flat surfaces are development calculations; one active-family direct execution independently verifies a bounded subset but not the entire five-scale preview. The silicon case study is a carefully bounded material-derived application: it authenticates inputs and producer outputs but lacks an independent reconstruction. None is a sealed public archival package.

Complete-scope execution and independent reconstruction are distinct evidence axes. Complete scope concerns whether the declared numerical inventory was evaluated; independence concerns whether separate implementation structure and arithmetic recover the result from authenticated inputs. Independent reconstruction of a subset does not verify the unexamined inventory, while a complete rerun of the producer does not become independent merely by covering every case.

Result class	Evidence type	Verification status	Claim permitted
Rank-two loss geometry	Analytic theorem	Proof under stated assumptions	Exact restricted-family identities
Dispersive rank-two endpoint	Sampled development calculation	Bounded independent subset only	Declared sampled cutoff comparison
Rank-10 gauge/locality	Authenticated producer calculation	Independent reconstruction pending	Source-specific qualitative mechanism
Literal/refitted shells	Exhaustive literal family and constructive searches	Producer checked; no global optimizer certificate	Observed witnesses and support tradeoffs
Frozen fidelity and band edges	Source-relative dense, Parseval, and local diagnostics	Mass conventions independently audited; scientific reconstruction pending	Internal fidelity of the frozen source models
DFT parent comparison	Separately retained parent evidence	Context and adequacy audit	Source inadequacy, not reduction fidelity or experiment

This layered status affects publication language, not the numerical values themselves. The correct response is to label each claim by its evidence owner and scope, rather than either suppressing useful development results or presenting them as publication-qualified validation.

8 Conclusions

The results below combine exact statements for a restricted rank-two family with bounded numerical demonstrations for dispersive and rank-10 representations. The material-derived numerical results remain subject to independent reconstruction. Within that evidence hierarchy, the combined theory and calculations establish six main conclusions.

1. Band equivalence does not determine a matrix representation or its hopping locality. Momentum-dependent unitary similarity can preserve every complete band while redistributing Fourier weight.

2. In a flat rank-two real-rotation family, the restricted-alignment threshold obeys an exact Bessel law, and its square equals the induced non-onsite hopping power.

3. In the controlled dispersive rank-two example, the spectrum-preserving representation change increases the first passing cutoff from 4 to 10 at the sampled band-error target $10^{-8}E_{\mathrm{ref}}$. This is a result for the declared attack, support family, inventory, and criterion, not a universal monotonic gauge–range law.

4. In producer-checked development calculations, the rank-10 silicon source exhibits the same qualitative mechanism: complete spectra remain invariant while alignment residuals, block power, and observed support-error frontiers change.

5. Literal shell deletion and bounded shell refitting produce different candidate geometries. Refitting creates genuine spectral–operator tradeoffs, but tight fidelity in the tested whole-block Bravais-shell family still requires near-full support.

6. Global fitting losses do not control dense interpolation or band-edge fidelity. Conduction masses, valence degeneracy, multiband curvature, and symmetry-equivalent gaps expose errors that global spectral RMS can hide.

For the studied source, H-shell-16 is the only frozen reduced witness that remains close across all tested internal channels, but it retains 371 of 383 representatives and is a near-full-support reduction rather than a compact model. H-shell-05 and H-shell-10 are not reliable band-edge reductions. Even H-shell-16 does not validate silicon physics because the full source fails separate DFT-parent criteria, and none of these silicon conclusions is publication-qualified independent reconstruction.

A stronger future program would independently reconstruct the silicon analyses, retain the same frozen models for a separately labeled DFT overlay, and construct an SOC spinor source only if physical hole or acceptor claims are required. Any prospective acceptance threshold must be declared before a future confirmation set is inspected.

A Derivation details for the rank-two loss geometry

Let a traceless Hermitian 2×2 matrix be written as $D = x\sigma_x + y\sigma_y + z\sigma_z$. Conjugation by the real rotation $R(\phi)$ leaves the σ_y component invariant and rotates (x, z) through angle 2ϕ. The weighted correlation with an attacked field therefore has the form

$$b_\phi = b_0 + c\cos 2\phi + d\sin 2\phi, \qquad b_0 = \sum_i \frac{\alpha_i}{E_{\mathrm{ref}}^2} y(q_i)^2 \geq 0. \tag{83}$$

The continuous maximum is

$$b_{\max} = b_0 + \sqrt{c^2 + d^2}, \qquad \mu_{\max} = b_{\max}/\kappa. \tag{84}$$

When $c^2 + d^2 > 0$, a canonical maximizing angle is $\phi_* = \frac{1}{2}\,\mathrm{atan2}(d, c)$ reduced to the half-open fundamental interval. When the harmonic amplitude vanishes, the optimum value remains defined but the angle is nonunique.

The spectral residual separates into a scalar trace shift and a traceless splitting change. Trace and traceless matrices are Frobenius orthogonal, which gives Eq. (23). The operator loss expands as

$$L_{\mathrm{op},\phi}^2 = s^2 + \kappa\lambda^2 - 2\lambda b_\phi + \kappa \tag{85}$$

$$= s^2 + \kappa(\lambda - \mu_\phi)^2 + \kappa(1 - \mu_\phi^2). \tag{86}$$

Because $b_{\max} \geq 0$, positive λ makes maximizing b_ϕ equivalent to minimizing the loss and the unconstrained scale is not clipped. For a different orbit with a negative maximal correlation, minimization over $\lambda \geq 0$ would instead stop at $\lambda = 0$.

For the residual–locality identity, now impose the additional theorem assumptions that D is real symmetric ($y = 0$), $U(q)$ is a real rotation, and either normalized continuum measure or uniform finite weights are used. These restrictions ensure that B_0 lies in the same two-dimensional real-symmetric coefficient plane as the scaled rotation orbit of D; Eq. (31) is not asserted for a general $y \neq 0$ Hermitian parent.

Use normalized Brillouin-zone measure dq on $[-1/2, 1/2]$, so that $\int_{-1/2}^{1/2} dq = 1$, and define

$$B_R = \int_{-1/2}^{1/2} e^{-2\pi i q R} B(q)\, dq. \tag{87}$$

With this convention, Parseval for the flat parent gives

$$\sum_R \|B_R\|_F^2 = \int_{-1/2}^{1/2} \|B(q)\|_F^2\, dq = \|D\|_F^2\,. \tag{88}$$

The zero block B_0 is a real traceless symmetric matrix in the same two-dimensional coefficient plane as the real-rotation orbit of D. Hence a real rotation $R(\phi_*)$ aligns $R(\phi_*)^{\mathsf{T}} D R(\phi_*)$ with B_0. Minimization over the positive scale then gives

$$\lambda_* = \frac{\|B_0\|_F}{\|D\|_F} = \mu_{\max}, \qquad \lambda_* R(\phi_*)^{\mathsf{T}} D R(\phi_*) = B_0, \tag{89}$$

with the zero-block case understood by the corresponding infimum. Thus the best onsite candidate reproduces the zero Fourier block and

$$\sum_{R \neq 0} \|B_R\|_F^2 = \|D\|_F^2 - \|B_0\|_F^2 = \|D\|_F^2\,(1 - \mu_{\max}^2). \tag{90}$$

For the flat family, $\kappa = \|D\|_F^2\,/(2E_{\mathrm{ref}}^2)$. Dividing the last identity by $2E_{\mathrm{ref}}^2$ therefore yields Eq. (31) with the same normalization used by Eq. (21). On a finite mesh the same argument uses $\alpha_i = 1/N$ and the normalized discrete transform, so the optimal constant is the unweighted zero Fourier block.

B Calculation inventory and evidence scope

Table 4 lists the retained calculation packages consumed or discussed by this synthesis. Inclusion in this table does not transfer evidence status between packages. In this internal lifecycle vocabulary, "unsealed" means not yet finalized as an immutable public archive, "provider eligibility" means only that a structured operator passed software-interface admission, and "typed verification" means that a schema-bound software check passed. None of these terms denotes material validation or publication acceptance.

Table 4: Calculation and evidence inventory.

Owner	Role in synthesis	Current boundary
Rank-two band-equivalence/locality owner	Exact rank-two geometry, smooth attack, dispersive range inflation, separated error channels	Analytic results plus development calculations; passing unsealed direct-v2 covers only its declared bounded aggregate.
Mathieu scalar-reduction owner	Scalar Fourier-reduction precursor	Separately retained and independently verified; not evidence for the rank-two headline.
PIAB1D spectral-reduction calibration	Separates continuum, finite-matrix, eigensolver, and invariant-subspace compression roles	Retained development calculation with typed verification and bounded formal crosswalk; not evidence for silicon accuracy.
QHO1D spatial-truncation calibration	Separates finite-domain closure, grid refinement, retained dimension, and common-coordinate pullback	Retained unsealed development calculation; observed finite-box/grid regimes are not rigorous error bounds or silicon evidence.
Construction-mesh recommendation	Freezes the $C6$ construction successor and unobserved 402-point boundary	No calculator execution; 402 parent values remain excluded.
Untruncated $C6$ characterization	Authenticates native source, reconstruction, valleys, curvatures, and locality inventory	Structurally verified source characterization; not material adequacy.
Revealed-408 development audit	DFT-parent source-adequacy audit	Valley positions pass; gap and almost all masses fail. Development oracle only.
Parent-validation preflight and revealed Davidson NSCF evidence	Prospective criteria and direct parent comparison	Existing authorized DFT evidence; no rerun performed for reduction fitting.
Accepted-S1 parent band-path evidence	Conventional L–Γ–X–Γ DFT diagram and source overlay	One authorized fixed-density QE 7.5 invocation; contextual path only, with no model fitting or global-extremum claim.
Operator qualification	Source reconstruction and provider eligibility	Structural/provider status, not scientific acceptance.
Frozen-gauge sensitivity diagnostic	Earlier optimizer/gauge sensitivity diagnostic	Neither compared operator accepted; supports caution about iteration-only extension.
Spatial-truncation design	Freezes native complete-shell hierarchy	Design only; selects no reduced radius.
Spatial-truncation evaluation	Evaluates every literal native support	Source-relative results; no accepted nontrivial shell.
Spatial-truncation disposition	Retains full support as the formal selected truncation	Selected $H_T = H_W$; zero selected truncation loss.

Owner	Role in synthesis	Current boundary
Gauge/alignment/locality analysis	Rank-10 internal gauge attack, alignment, finite Fourier, locality, support curves	Producer checked, not independently reconstructed.
Literal two-loss exploration	Literal two-loss witness surface	Post-hoc descriptive thresholds; full support excluded from reduced feasibility.
Refitted shell-witness analysis	Bounded shell-scale witnesses and fitted surface	Constructive multistart evidence; no global infeasibility certificate.
Wannier-only fidelity ladder	Parseval, dense meshes, conduction valleys, gaps, Hessians, masses	Frozen source-relative models; no DFT validation.
Comprehensive band-edge analysis	Full electron tensors, valence cluster, multiband quadratic operator, direct gaps, bandwidths	Producer checked non-SOC development analysis; no HH/LH/SO claim.
Independent numerical-convention audit	Recomputes six-valley mass conventions and the exact size-six cyclic/ordinary mode ledger	Separate arithmetic passes; does not reconstruct Hamiltonian fields, valleys, Hessians, or removed-mode coefficients.
Slater–Koster common-space gate	Attempted common-space gate for joint SK fitting	Stopped at rank 9 at Gamma and one additional point; no fit or parameter vector.
Hopping Pareto synthesis	Protocol-level composition of channel-specific frontiers	Unresolved development evaluation; selects no support.
Historical tutorial silicon records	Software, parsing, and bounded same-frame decomposition controls	Not production silicon evidence and not the source of the rank-10 results.

C Consolidated numerical tables

Narrative values are rounded for readability. The tables below retain the reported precision needed to identify the machine-readable records; their extra digits are not uncertainty estimates.

C.1 Rank-two endpoint range decisions

Band-error target	Reference cutoff	Comparison cutoff	Inflation
$10^{-6}E_{\mathrm{ref}}$	3	8	5
$10^{-8}E_{\mathrm{ref}}$	4	10	6
$10^{-10}E_{\mathrm{ref}}$	6	12	6
$10^{-12}E_{\mathrm{ref}}$	7	14	7

C.2 Frozen silicon fidelity ladder

Model	Parseval operator RMS (eV)	32^3 spectral RMS (eV)	Gap change (meV)	Max mass change
H-shell-05	0.260036	0.181957	-65.77	67.7%
H-shell-10	0.059558	0.027851	-34.09	53.2%
H-shell-16	0.001747	0.000657	-0.503	2.54%

C.3 Valence and symmetry fidelity

Model	Gamma splitting (meV)	Quadratic error	Max channel-mass change	X/L gap spread (meV)
Full source	0.02495	—	—	0.00366 / 0.0239
H-shell-05	102.175	9.54%	38.9%	150.1 / 68.7
H-shell-10	54.806	5.88%	19.7%	36.7 / 63.9
H-shell-16	2.496	0.279%	4.53%	0.427 / 2.97

D Analysis procedures and reproducibility record

This appendix records the analysis order in sufficient detail to prevent a later narrative rewrite from collapsing distinct operations. It is not by itself a complete reproduction package and does not replace the authenticated source data, executable environment, dependency lock, retained requests, producer scripts, or an independent reconstruction path.

D.1 Rank-10 gauge analysis

1. Authenticate the source lattice, rank, units, construction inventory, evaluation inventory, and represented matrices.

2. Evaluate the frozen rank-10 Givens circuit and its known inverse on each requested reciprocal point.

3. Construct the attacked field by pointwise unitary similarity. Verify unitarity, Hermiticity, trace invariance, and complete-spectrum invariance before alignment.

4. Evaluate identity, one-global-unitary, pointwise-polar, and known-inverse alignments. Keep restricted residuals distinct from oracle residuals.

5. On the centered 6^3 construction tensor only, compute finite-cyclic Fourier blocks. Convert them explicitly to ordinary representatives and record power removed by conversion.

6. Reconstruct both finite fields and evaluate interpolation losses on the ordered 408-point inventory. Do not attribute these finite-route errors to the exact attack.

7. Apply the same complete support inventory to both finite representations. Evaluate spectral and operator consequences at every support and extract discrete non-dominated points without interpolating fractional supports.

The retained reconstruction ledger is summarized in Table 5. All listed files are relative to `calculations/Si/bulk/analysis/c6-b20-w21-gauge-alignment-locality/`. The machine-readable files, rather than the prose description, own the exact arrays and ordering.

D.2 Literal and refitted native-shell analyses

The literal analysis loops over H-shell-00 through H-shell-17, reconstructs each native Wigner–Seitz model on the 408-point inventory, and computes τ_1 and τ_2 with identity alignment. Minimum-support tables are then constructed over an explicit grid of post-hoc threshold pairs. H-shell-17 is retained as a control but removed from the proper-reduction feasibility surface.

For refitted shells, each support S defines a coefficient vector $\boldsymbol{\alpha} \in [0,2]^{|S|}$. The producer precomputes shell-resolved matrices and analytic derivatives of the scalarized loss. Seventeen scalarization weights and three deterministic starts produce at most 51 optimization attempts per support. Converged solutions are deduplicated in loss/parameter space, authenticated, and retained as constructive witnesses. Literal points remain visible as a separate family.

The gradient audit compares analytic and finite-difference derivatives over retained coordinates. The maximum absolute difference is 9.7361×10^{-10} and the maximum scaled difference is $4.0729 \times$

Table 5: Machine-readable reconstruction ledger for the rank-10 gauge analysis.

Stage	Retained artifact	Auditable content
Source and coordinate contract	`request.json`	Source hashes; native and centered construction coordinates; zero offset; ordered 408-point role and weights; attack identity; tolerance policy
Producer implementation	`run_analysis.py`	Native HR/Wigner–Seitz reconstruction; centered permutation; transform, conversion, support, and error-channel algorithms
Construction and attack result	`result.json`	`operator`, `attack`, `alignments`, and `producer_checks`; complete spectrum and Hermiticity checks
Ordinary displacement inventory	`result.json:displacement_power`	Integer modes, metric-shell labels, physical radii, reference and attacked block power
Support and Fourier channels	`result.json:alignments[*].shells`	Exact support inventories, reconstruction, conversion/interpolation/truncation channels, and observed non-dominated records
Evaluation ordering	`result.json:evaluation_spectra`	Ordered point indices, fractional coordinates, source and attacked spectra, and separately labeled parent overlay

10^{-11} for the audited cases. This verifies the implemented derivative calculation; it does not establish global optimization.

D.3 Frozen fidelity escalation

The fidelity ladder is intentionally one way. Three witness identities are copied from the completed refitted study and frozen. Subsequent stages may evaluate but not alter their shell scales.

1. Recompute the 408-point spectral and operator RMS losses from the frozen models.
2. Compute exact integrated operator RMS by Parseval from the unique Fourier coefficients.
3. Evaluate sorted spectra on independent 12^3, 20^3, and 32^3 meshes.
4. Partition the 32^3 mesh into six dominant signed-Cartesian-axis sectors. For each sector, retain its eight lowest coarse points plus the source valley coordinate, minimize zero-based band 4 from every start in a Cartesian displacement box $[-0.12, 0.12]^3$ Å^{-1} with at most 240 L-BFGS-B iterations, function tolerance 10^{-14}, and gradient tolerance 10^{-10}, reject candidates whose final dominant signed-axis label differs from the requested sector, and retain the lowest successful matching candidate as the identified sector minimum.
5. Evaluate gaps, valley displacement, Hessians, and principal masses without refitting.
6. Promote the same models into the comprehensive band-edge analysis.

A preliminary valley search that encountered a 0.08 Å^{-1} search boundary was not retained as the final result. The final request and producer record the enlarged search and multi-start sector logic. This change modifies the evaluation method, not the frozen Hamiltonians.

D.4 Transported valence quadratic analysis

For each model and displaced momentum $\mathbf{q}$:

1. extract the three-dimensional upper-valence eigenspace;
2. form its overlap with the model's Gamma cluster;
3. compute the polar unitary and transport the displaced represented matrix into the Gamma-cluster coordinates;
4. average the transported $+\mathbf{q}$ and $-\mathbf{q}$ matrices to remove odd numerical terms;
5. fit the six Hermitian quadratic coefficient matrices in Eq. (78); and
6. align the candidate coefficient field to the source cluster before reporting the joint Frobenius error.

This algorithm never assigns an individual band identity at the degenerate point. The reported channel masses are derived after direction contraction and diagonalization of the fitted quadratic matrix.

E Error-channel definitions and interpretation

Channel	Primary comparison	Interpretation and exclusions
Direct-parent adequacy	DFT parent versus full Wannier source	Tests the represented source against its parent calculation. Not a truncation or gauge loss. Current gap/mass targets fail.
Exact gauge equivalence	Full attacked field versus full source spectrum	Pointwise unitary identity. Does not imply identity/global matrix alignment or equal Fourier support.
Restricted frame residual	Matrix fields after identity or one-global-unitary alignment	Measures mismatch under the declared smaller alignment family. Not physical inequivalence of the complete gauge-related fields.
Cyclic representation	Centered finite transform versus its exact sampled field	Finite-discrete representation channel. Must not be silently substituted for native Wigner–Seitz HR data.
Conversion	Finite-cyclic representatives versus ordinary lattice representatives	For the size-six route, removes negative-Nyquist-boundary modes and maps every retained mode identically. Distinct from deliberate support truncation.
Interpolation	Frozen finite field on evaluation points not identical to construction points	Measures reconstruction between construction samples. Not a gauge error.
Literal truncation	Deleted native shell model versus full source	Coefficients inside support remain unchanged. Measures deletion only.
Refitted reduction	Bounded shell-scaled model versus full source	Includes compensation by retained shells. Does not have Slater–Koster orbital semantics.
Dense spectral fidelity	Frozen model versus source on N^3 meshes	Broader source-relative interpolation check. Not DFT validation.

Parseval operator fidelity	Exact coefficient difference over the normalized Brillouin zone	Integrated same-gauge operator RMS for the finite Fourier series.
Conduction feature fidelity	Source versus frozen model valleys, Hessians, and masses	Local derivative check. Curvature masses are not transport simulations.
Valence multiband fidelity	Aligned three-band quadratic coefficient fields	Degeneracy-aware non-SOC comparison. No HH/LH/split-off assignment.
Total finite-model consequence	Final reduced finite model versus declared source	May contain several prior stages but is not decomposed by subtracting scalar errors unless an explicit identity permits it.

F Artifact identification and reproducibility anchors

The following hashes identify central retained outputs used in drafting this synthesis. They are audit anchors, not reproducibility artifacts or publication DOIs. A public archive must preserve the referenced files themselves together with manifests, source identities, execution records, dependency resolution, and the relationships among those objects; uploading only this PDF and a list of digests would not reproduce the results.

Artifact	SHA-256
Literal shell `result.json`	`2933765cb26d4aee05dca69fd73ec45f2ab26a32a145ebf9e8af80db0fdaa503`
Literal witness surface PDF	`2184e4b5b58040aaa48adfd8926384ea3fb49ee13618d5243e5fb87e3025beec`
Refitted shell `result.json`	`8a65cf3fbe3b08e8f7b71a802b8400515e1a0a514efd031e14783d6973a65664`
Refitted witness surface PDF	`cabe28a205c6dfcdd955786b096011bc98c23a0f3fbb35bea8b11e1ec840a502`
Fidelity ladder `result-v2.json`	`781c29c9512c643bbf6e23a6d524a2466caaba501dc9c7323954ba7782296fdc`
Fidelity ladder figure	`55551ae572b1f6dc4f24971e758aed1b9a720c67ae967e328d54f71e23793b09`
Source-relative feature figure	`0520aaf5170a640880a220ab9f28eed45e579774502fa63e4f41fec16b2a153e`
Comprehensive band-edge result	`533dfcbed4883793bb84d1a13d1a8f62f7bb7f99b02d3ab07bb3ea99f78b0764`
Comprehensive multiband result	`bb7be7af470c521a69897fd910fe476dce40eee1382ff8e46e6040b851718ea6`
Comprehensive manifest	`226d654c592052a688b954009cea96b7e2b4e5c7b953d4614ca02cafa0bdf614`
Independent convention-audit result	`56c15bd597056cb1a607f2cd88100bb10da94af55d29f6606d868fb353d513cc`
Cyclic-to-ordinary conversion ledger	`36f4e06e58081f2b6e13e42b77bf6b5b6f45aa337c2983bbb02596b7c932e036`
Convention-audit manifest	`7b85ebd9d9cfd534e42519e491344ad561aa03c99bc71d3491ac9105142f79ec`
Parent band-path `band-data.json`	`e4a4c04eabb4cdc8f0dfef2e4b175c90ab3529b7406fbc29b2aa1aee148b5bcd`

Parent DFT band-diagram PDF	cd96619a6c6082e63bda4ea53f36ee737ab07615 1a9097989cc8642f25940bcd
Narrative technical report v2 PDF	72f6850bcf71d9855d33888c3ef580ca160f0836 c8e08bb0622fe6f6e966ea69
Rendered provenance-plan manuscript PDF	bf8d4c680cf489134973539e34992df90e13d866 84afc0bc9143f432e0091bd5

The calculation manifests, not this table, remain authoritative about complete artifact membership. A future rebuild of this manuscript may change its PDF hash without changing any underlying calculation result.

G Scope of interpretation and remaining limitations

G.1 Claims supported at the present scope

1. Exact pointwise band equivalence does not determine hopping locality.
2. The rank-two flat-parent threshold and residual–tail identities hold under the displayed assumptions.
3. The declared dispersive development calculation exhibits positive sampled range inflation over most of its amplitude/scale inventory.
4. The frozen rank-10 attack preserves the complete represented spectrum while changing restricted alignment and finite-block locality.
5. Literal and refitted native-shell families have different two-loss geometry.
6. The 408-point witness losses understate broader frozen-model errors.
7. Moderate shell models do not preserve conduction or valence band-edge structure relative to the full source.

G.2 Claims explicitly not supported

1. The source is not validated as an adequate silicon Hamiltonian.
2. No experimental silicon agreement is established.
3. No universal hopping-range law or universal spectral/operator conversion is claimed.
4. No physical HH, LH, split-off, or acceptor result is claimed.
5. No Slater–Koster incompatibility theorem is claimed; the attempted joint fit was blocked by a common-space rank gate.
6. Gray witness cells are not infeasibility certificates.
7. Development plots and producer checks are not independent publication evidence.
8. The frozen 402-point confirmation values were neither accessed nor inferred.

G.3 Before any submission or promotion

A publication-qualified version would require a frozen claim set, reviewed numerical controls, complete-scope retained execution where applicable, independent reconstruction of the silicon analyses, final authorship and acknowledgments, a public archival record, and explicit authorization for submission. The retained provenance distinguishes the earlier 0.5226105 eV lower-energy-frozen $4^3/12$-band candidate from the present 0.6892447 eV $C6$–$B20$–$W21$ candidate; these are different Hamiltonians compared with the same revealed DFT parent, not conflicting estimates for one source. If physical hole claims are retained, a separate SOC source and state-space contract are required. If DFT-relative overlays are added, they must remain separate from Wannier-relative compression losses and must not be used to refit the already frozen models.

Data and code availability

The working repository stores the rank-two calculation package under `calculations/periodic1d/rank2/band-equivalence-versus-hopping-locality/` and the silicon analysis packages under `calculations/Si/bulk/analysis/`. Each package retains requests, source identities, producers, machine-readable results, figures, manifests, and execution records. The comprehensive band-edge result and its multiband amendment are under `c6-b20-w21-wannier-only-comprehensive-band-edge/`. The narrative technical report is under `calculations/Si/bulk/technical-report/v2/`.

This manuscript is a synthesis surface only. It does not duplicate or replace the calculation records, and no numerical value is owned solely by the prose above. At the time of this working revision there is no public repository URL or archival DOI that contains the complete authenticated source data and reconstruction environment. Consequently the local paths and SHA-256 anchors are audit aids, not an externally executable reproduction package. Public deposition and DOI citation remain mandatory before submission.

Author contributions

E.J.M.R. developed the formulation, implemented and executed the reported development analyses, interpreted the results, and wrote the manuscript. Authorship and contribution statements remain subject to final review before submission.

Conflict of interest

The author declares no conflict of interest in this working manuscript.

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